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Cities Without Skylines: Worldwide Building-Height Gaps and their Possible Determinants and Implications☆

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ABSTRACT

There is a large literature on U.S. cities measuring the extent and stringency of land-use regulations and how regulatory and geographical constraints affect important outcomes such as housing prices and economic growth. This paper is the first to study the global extent and impact of regulatory and other constraints by estimating what we call *building-height gaps*. Using a novel data set on the year of construction and heights of tall buildings around the world, we compare the total height of a country's stock of tall buildings to what the total height would have been if supply was more elastic, based on parameters from a benchmark set of countries. These gaps are larger for richer countries and for residential buildings than for commercial buildings in such countries. The gaps are driven by under-building in central areas of larger cities. These gaps are not compensated by tall building construction in peripheral areas of cities or less stringent limits on outward expansion beyond the existing boundaries of the cities. Countries with older, historic structures have larger gaps, likely due to more stringent height regulations and dispersed ownership that inhibits land assembly. Lastly, the gaps correlate strongly with international measures of housing prices, sprawl, congestion, and pollution.

Today, the majority of the world's population lives in cities, and the urban population will continue to grow over the rest of the century (World Bank, 2009; UN-Habitat, 2020). Cities throughout the world must expand their stock of real estate in order to accommodate urban growth (Glaeser, 2011; Romer, 2020). But in many countries, housing prices are growing more rapidly than incomes (Knoll et al., 2017), which may be partly caused by physical and regulatory barriers reducing housing supply (Glaeser and Gyourko, 2003; Glaeser et al., 2005; 2006; Saiz, 2010; Gyourko and Molloy, 2015; Harari, 2020).

Regulatory and other constraints on development not only have impacts today but may generate significant effects well into the future given the durable nature of real estate (Glaeser and Gyourko, 2005; 2018; Hsieh and Moretti, 2019). Higher housing prices can also lead to more exclusionary cities (Gyourko et al., 2013; Rosenthal and Ross, 2015). While the extent and impact of regulatory or geographical constraints has been studied for the U.S. (Ihlanfeldt, 2007; Saks, 2008; Saiz, 2010; Brueckner and Singh, 2020), there are no such studies for the whole world. Offering such a study is the aim of this paper.

More generally, across countries, some cities appear more willing than others to construct tall buildings in their central areas as a way to accommodate economic and population growth. Cities in China embrace tall buildings (Barr and Luo, 2021), whereas cities in India have draconian height restrictions (Brueckner and Sridhar, 2012). Some cities in Europe seem to represent an intermediate case, with tall buildings emerging in London (Cheshire and Dericks, 2020) and Frankfurt, al-

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though other cities, especially in southern Europe, have few tall buildings. These patterns raise three questions. First, how many tall buildings are "missing" in each country and the world overall? Second, what could be the economic and socio-cultural determinants of these building-height gaps? Third, what could be the economic and environmental consequences of these gaps?

Our approach makes use of a remarkable data set that inventories all the world's *tall buildings* (buildings above 80 meters), with information on their year of construction and height. Using a set of more supply-elastic countries as a benchmark, we ask whether the stock of tall buildings in a country outside this set is smaller than expected given the country's characteristics. More specifically, we start by running a panel regression (1950-2020) relating a measure of the tall building stock per capita in the set of identified benchmark countries to two variables suggested by the standard urban model (the main model used to study urban land use): income and agricultural land rent. Then, for countries outside the benchmark group, we plug values for these variables into the estimated equation, yielding a predicted size for the tall building stock per capita if the country's supply elasticity matched those in the benchmark group. The difference between the prediction and the country's actual stock is the *building-height gap*.

We find that the world might have a shortage of tall building as large as the entire existing stock. In other words, due to regulation and other impediments, the stock of tall buildings may fall short of the stock that would exist under laissez-faire practices by about 6,000 Empire State Buildings. The gaps in tall building stocks are larger for richer countries, especially European countries. Globally, the gaps are larger for office towers, due to various world regions having disproportionately large residential tall-building stocks (e.g., the MENA countries). However, in developed countries, the gaps are larger for residential buildings, suggesting that these countries are more open to creating jobs than to receiving residents.

Conditional on income, the gaps are strongly and positively correlated with housing prices, sprawl, road congestion, and pollution, implying that they might have important global economic and environmental consequences. In particular, such gaps might explain one fifth of the global house price boom observed since 1950 (Knoll et al., 2017). The gaps then appear to constrain the growth of large cities and their central areas. However, we do not find that countries with larger building-height gaps "compensate" their urban residents (i) by having more lenient height restrictions outside their cities' central areas; (ii) by having more lenient sprawl policies; (iii) by subsidizing transport; or (iv) by providing public housing. Given the lack of global historical data on non-tall buildings, we are unable to assess whether higher-gap countries have smaller gaps in this other dimension. However, the fact that housing prices are higher in higher-gap countries suggests that any compensation via the non-tall sector may not be sufficient.

Regarding the nature of these gaps, we find no evidence that they are caused by geological, topographical, or seismic conditions, or differences in construction costs. However, countries with more "historical" cities (cities with long histories) have higher gaps. Evidently, the gaps in these countries arise from regulations designed to protect valuable historical urban areas, motivated by cultural considerations or a desire to foster tourism (Viladecans-Marsal and Garcia-Lopez, 2021). However, countries may also adopt stringent regulations for reasons unrelated to historical preservation.[1]

While we believe our results are novel, one limitation is that the gaps are directly estimated at the *country* level. Ideally, we would have liked to use *city*-level data to estimate city-level gaps and then obtain country-level gaps from them. To do so would require a panel data set for as many cities across the world as possible, and which would include city population, GDP, and land rent over several decades – giving us a long enough time period to run panel regressions with city fixed effects for the proper estimation of the gaps (preferably 1950-2020). Unfortunately, historical panel data on city incomes and land rent does not exist except in a few countries. As well, usable night lights data as a proxy for income are relatively limited, and non-existent for the years before 1992. Converting night lights to useful data has measurement problems, and night lights capture other urban phenomena not related to income. In addition, consistent historical agricultural land rent data do not exist for the "edge" of each city of the world.

Nevertheless, we will show that our country-level gaps are correlated with city-level gaps estimated using city-level panel regressions involving city-level data where possible. More generally, we will discuss how ignoring the city size distribution of each country should not affect our results. Finally, within-country analysis also misses the fact that constraints on supply in specific cities may be circumvented by migration to less constrained cities in the same country, hence the need for a country-level analysis.

Our paper makes several contributions to the literature. To our knowledge, the above results do not appear in any other study, given the lack of historical international data on tall building construction, which only recently has become available. The paper constitutes one of the first global studies on the economics of skyscrapers (see Ahlfeldt and Barr (2022) for a survey). Other studies typically focus on the U.S. only (e.g., Ahlfeldt and McMillen, 2018; Liu et al., 2018, 2020; Rosenthal and Strange, 2008; Liu et al., 2022).[2] Next, there are almost no studies on how building-height regulation varies across countries, and little is known about the determinants and effects of building-height variation globally. Finally, our method has the advantage of being replicable for any context. So far, only a few other studies have implemented this type of indirect method to measure the stringency of regulation, defined as the degree to which regulations and other frictions cause development decisions to differ from free-market outcomes.[3]

The extent of land-use regulation in U.S. cities has been captured through a number of regulatory surveys, which present local government officials with a long list of different types of potential regulations, asking which ones are used in their community. While some surveys focus on specific localities or specific states (e.g., Glickfeld and Levine, 1992; Ihlanfeldt, 2007; Jackson, 2018), Gyourko et al. (2008) and Gyourko et al. (2019) carry out more ambitious national surveys. These detailed studies then use the responses from local communities to compute a regulatory index for individual cities or states.

Despite their impressive scope, these surveys lack coverage of certain important regulations. For example, Gyourko et al. (2008) – on which the *Wharton Residential Land Use Regulation Index* is based – and Gyourko et al. (2019) do not measure height restrictions in the central areas of cities. Rather, they focus on frictions due to the permitting process and on other regulations that are more likely to impact low-rise dwellings in suburban parts of the city. Another issue is how to aggregate this community-level information into reliable measures for cities, states or even the country. The survey method is also expensive and thus cannot be used to compare regulations across countries. That is why *no* global measures of land-use regulations exist. Finally, such measures cannot tell us if the regulations are binding given economic conditions.

In contrast, our method of analysis is similar to that followed in the health sciences literature. Currently, two methods exist to measure and compare domestic and international mortality rates due to an environmental phenomenon (e.g., a pandemic). National agencies can compile

[1] In contrast to the U.S. literature (see Rosenthal and Ross (2015) for a discussion), we do not find that the gaps are driven by the homevoter hypothesis. The hypothesis says that homeowners, worried that increases in supply reduce property values, lobby cities to impose regulations (Fischel, 2001).

[2] For other research on building heights, see, for example, Barr (2016), Barr and Cohen (2014), Barr (2012), Brueckner and Sridhar (2012b), Barr (2010) and Bertaud and Brueckner (2005a).

[3] See Bertaud and Malpezzi (2001), Glaeser et al. (2005), Turner et al. (2014), Brueckner et al. (2017a), Brueckner and Singh (2018) and Albouy and Ehrlich (2018) for important works on land-use regulation.

phenomenon-related mortality information obtained from each jurisdiction, an imperfect process if jurisdictions and countries have different measurement abilities. An alternative indirect method is to perform simple regression analyses to estimate "excess mortality rates" for all jurisdictions and countries.[4] Our method is analogous to the latter approach, except that we are measuring a shortage of tall buildings. While our approach might appear simple, it has the benefit of being feasible and, as our evidence suggests, possibly reliable. However, given the lack of data, one limitation is that we cannot attribute the gaps to land-use regulations only.

Lastly, our analysis is mostly a predictive exercise, which makes causality less of an issue. It is not possible to find an instrument, or more generally an identification strategy, that would explain income and land rent (our right-hand side variables) without also directly impacting the stock of tall buildings (at the country or city level). Yet, we discuss a series of robustness checks that aim to give greater confidence in the results.

The plan of the paper is as follows. Section 1 provides the conceptual foundations for the panel regressions that we run. Section 2 discusses the data and background. Section 3 discusses the international regression results and the associated building-height gaps. Sections 4 and 5 examine the possible determinants and effects of the gaps, respectively. Section 6 reports a series of robustness checks. Section 7 concludes.

1. Conceptual Framework for the Panel Regressions

This section provides the conceptual foundations for our panel regressions. We will use the coefficients estimated for the benchmark countries to compare the total height of each country's stock of tall buildings to what the total height could have been if supply was more elastic, hence the need to itemize the factors to be considered econometrically. We first discuss the Standard Urban Model (SUM) and then additional elements.

1.1. The Standard Urban Model (SUM)

The SUM, as exposited by Brueckner (1987) and Duranton and Puga (2015), depicts the determination of building heights, measured by the quantity of floor space per unit lot area. In the model, consumers value access to jobs in the city center, which leads to both higher housing prices and higher land rents near the center. Faced with expensive land, developers construct taller buildings near the center to limit use of the expensive land input. In the equilibrium of a closed city (where population is fixed), building heights depend on the city's characteristics, which include population, per capita income, commuting cost per mile, and the agricultural rent for the land surrounding the city. A higher population or agricultural rent raises building heights throughout the city. To accommodate the greater demand from a larger population, the city must be denser, with taller buildings. By increasing the cost of rural-to-urban land conversion and thus making the city more compact, a higher agricultural rent also generates taller buildings.

A higher income causes urban decentralization as residents find the cheap suburbs more attractive for the bigger dwellings they now prefer. This demand shift tends to raise building heights in the suburbs while decreasing them near the city center, yielding a spatially complex income effect (higher commuting costs lead to the opposite impact).

The fact that countries become more urbanized as they get richer, however, has implications for spatial structure beyond the static version of the SUM. This tendency implies that urban populations in a country tend to rise with the general income level, suggesting that income and population increase in step with one another across countries. Allowing these variables to change together, the result is a tendency for building heights to rise uniformly across space within cities as country income increases. In particular, with cities tending to be bigger in high-income countries, the positive population-induced effect on heights offsets the negative height effect in the central city due to income-induced decentralization. As a result, we would expect that buildings in the richer country's cities should be taller in the center and the suburbs (Ahlfeldt and Barr, 2022).

Commuting cost per mile, another key SUM parameter, is difficult to measure. International data limitations prevent us from quantifying urban highway investment, for example, which lowers these costs, likely leading to decentralization (see Baum-Snow (2007, 2020)) and thus shorter central-city buildings. However, as in the case of income, highway investment and urbanization are positively correlated, with the resulting higher city populations likely to offset the negative effect of better highways on central building heights, making buildings taller throughout the city (Ahlfeldt and Barr, 2022).

Thus, we hypothesize that income, urban highway investment, and city populations all move in step as urbanization proceeds, and that the population effect dominates in determining the heights of buildings. Treating income as a proxy for urbanization, we then expect its effect on heights to be positive. In a regression relating building heights to a country's income and agricultural rent, the income and agricultural-rent coefficients should therefore be positive.

1.2. Additional Elements

Considering elements outside the SUM, we can assume that as cities achieve higher incomes, they add more amenities, which tend to be downtown. By making city centers more desirable as income rises, amenities can help to reverse the income-driven tendency toward decentralization, strengthening the tendency of building heights to rise at all locations when income increases.

If larger cities have greater traffic congestion, which tends to raise commuting costs, then the decentralizing effect of a better transportation infrastructure may be offset. Congestion thus reduces the downward pressure of highway investment in large cities on central heights, making the positive pure population effect more likely to dominate.

Finally, if we add business land-use to the residential use discussed above, the urbanization-driven association between country income and heights is likely to be magnified. High country incomes tend to be associated with the presence of service sector firms, which may value being located in city centers in order to reduce the costs of accessing inputs and consumers and to benefit from knowledge spillovers (Moretti, 2012). Thus, higher incomes should increase the demand for office space in city centers and cause tall building construction there.

In summary, the theories of urban spatial structure suggest a reduced-form empirical specification in which a measure of the stock of tall buildings is regressed on income and agricultural rent, with the expected coefficient signs both being positive. The analysis, however, applies to a city with perfectly malleable capital, where building heights adjust immediately to reflect current conditions. In reality, tall buildings are long-lived, having been built in response to current conditions at the time of construction and lasting decades. In a model recognizing this longevity, income and agricultural rent would only determine the *increment to the tall-building stock* through an effect on new construction. The existing tall-building stock would also be a determinant of this increment. These considerations suggest a regression with the existing tall-building stock as dependent variable and the lagged stock, income, and agricultural rent as covariates.

Finally, the exact definition of the tall-building stock measure requires discussion. As mentioned above, the stock measure is a weighted one, with each tall building weighted by its height. Since the stock is measured for the entire country, not for individual cities, it is appropriate to divide the weighted stock variable by the size of the country's urban population. The dependent variable is thus the country's height-

[4] See, for example, Dushoff et al. (2005); WHO (2018); Kiang and Buckee (2020); Beaney et al. (2020).

weighted stock of tall buildings *per urban capita*, which we call "urban height density."

2. Data and Background

To estimate our baseline model, we collected data on building heights, urban populations, urban per capita incomes, and agricultural land rents for as many countries and years as possible. Our main sample comprises 158 countries decadally from 1950 to 2017.

Building Heights. The Council on Tall Buildings and Urban Habitat (CTBUH) maintains an online database of all *tall buildings* in the world.[5] For each building, we extracted information on the building's height, year of construction, usage, and other characteristics. As described in Web Appx. Section 1, the database mostly captures buildings above 80 meters. Since some countries have no such buildings, in order to avoid having their stock of heights equal to 0 when using logs, we consider for all the countries in our sample buildings above 80 meters as well as their 10 tallest buildings even if some of them are below 80 meters. In the end, we use 16,369 tall buildings. Below, we will discuss how measurement error in building height stocks could impact our results.[6]

Urban Population. The United Nations (2018) gives the urban population of each country every 5 years from 1950 to 2020. We interpolate the data for intermediate years.

National Income. Our main source is Maddison (2008), where we obtain per capita GDP for each country annually from 1950 to 2008 (in 1990 Geary-Khamis dollars, which is equivalent to PPP and constant international 1990 $). We use per capita GDP growth rates from the World Bank (2018) to reconstruct per capita GDP from 2008 to 2017.

Agricultural Land Rent. We estimate a country's agricultural land rent by dividing agricultural GDP by the total land area. We use as our main source FAO (2018), which shows the agricultural GDP shares for many countries annually from 1960 to 2017. For country-years that are still missing, we use additional sources and interpolations as needed (again Appx. Section 1). We use total land area as the divisor instead of agricultural land area because the latter area is missing for almost all countries before 1960.[7] In addition, a significant share of non-agricultural and non-urban land can potentially be used for agricultural purposes or be converted into urban land. Below, we will discuss how measurement error in agricultural land rent could impact our results.

Urban Per Capita Income. Knowing for each country-year total GDP (PPP and constant international $) and the agricultural GDP share, we can reconstruct urban GDP, which we proxy by non-agricultural GDP. Knowing urban GDP, we can then reconstruct urban per capita income as urban GDP divided by urban population.

Urban Height Density. When logged, this is our dependent variable, equal to the sum of the heights of the country's tall buildings in a given year divided by the urban population for that year. We are also able to distinguish residential and office buildings.

Other Variables. We know from the World Bank the income group of each country in 2017 ("low," "lower-middle," "upper-middle" or "high income"). High income countries are viewed as developed. From The Economist (2018), we know whether each country was democratic at any point in the 2006-2017 period (data not available before).[8]

Descriptive Patterns. Figure 1(a) shows the evolution across time of the urban height-density measure for the U.S. along with the evolution of the same measure summed across all the world's cities. As can be seen, the U.S. contained virtually all of world's tall buildings up to 1950, with the two curves diverging thereafter. In recent years, the tall-building stock outside the U.S. has grown rapidly. Figure 1(b) shows the world evolution of the total stock of heights separately for residential buildings and office buildings from 1920 to 2017. As we can see, it is only after 2000 that tall residential buildings were built at a faster pace than office buildings. Circa 2017, residential buildings and office buildings each contribute about half of the total stock of heights in the world.

Figure 2 shows the relationship between the country-level log of urban height density in 2017 and the log of national GDP per capita for that year. As expected, the relationship is positive, with a strongly significant slope coefficient of 1.35*** and an R^2 of 0.52 (1.41*** and 0.67 if using urban population weights). Countries above (below) the dashed line have more (fewer) tall buildings than expected based on their income.

Finally, Figure 3 shows the absolute change in the total sum of tall building heights (km) experienced by each world city between 1950 and 2020 (proxied by 2017). As we will describe in Section 6, the cities correspond to 11,719 urban agglomerations in the *Global Human Settlement* database of the European Commission (2018).

3. The Gaps

Table 1 shows panel regressions for the years 1950, 1960, 1970, 1980, 1990, 2000, 2010, and 2017 (henceforth, "2020"). The explanatory variables are: log per capita urban GDP (LUPCGDP), the log of agricultural land rent (LAGRENT), and the lag of the dependent variable, log urban height density (LUHTDENS). We include country and year fixed effects (standard errors clustered at the country level). More precisely, for countries c and years t, the model is:

$$\text{LUHTDENS}_{c,t} = \alpha + \beta\text{LUPCGDP}_{c,t} + \gamma\text{LAGRENT}_{c,t} + \delta\text{LUHTDENS}_{c,t-10} + \theta_c + \kappa_t + \mu_{c,t}.$$

Column (1) includes 158 countries. The GDP coefficient is positive and significant, as is the lagged height-density coefficient. The coefficient on this variable is less than one, indicating that an increase in the lagged tall-building stock is associated with a less than one-for-one increase in the current stock, given that the increase in the prior stock depresses new construction. The land-rent coefficient while positive, is not significant.

Column (2) restricts the sample to 73 countries with a positive residual in a 2017 regression that relates LUHTDENS to LUPCGDP and LAGRENT. These are countries where the tall-building stock is higher than could be expected today given the magnitudes of the covariates, a simple way to select the benchmark countries. Naturally, the GDP coefficient is larger than in col. (1). The coefficient of agricultural rent becomes positive and significant. Logically, for 85 countries with a negative residual (col. (3)), the coefficient of GDP is almost twice as small, and the coefficient of agricultural land rent is nil.

Column (4) restricts the sample to 14 democratic upper-middle (henceforth, "UM") or high ("H") income countries whose residual is above the 75th percentile value. We restrict this benchmark sample to more-democratic and more-developed countries because market forces are less free to operate in other countries. Column (5) focuses on 8 H countries: Australia, Canada, Hong Kong, Israel, the Netherlands, Singapore, South Korea, and Uruguay. The 6 UM countries excluded are Brazil, the Dominican Republic, Macedonia, Malaysia, Panama and Thailand.[9] While the selection of these countries might seem arbitrary,

[5] The online database can be found here: http://www.skyscrapercenter.com/. As one example, the webpage for the Burj Khalifa is found here: http://www.skyscrapercenter.com/building/burj-khalifa.

[6] According their website, the data have been "collected by the Council for more than 40 years [...] The Council relies on its extensive member network [of academics, land developers, architectural firms, builders, city administrations, and banks] to maintain" the database with the help of "an Editorial Board."

[7] From FAO (2018), we know total land area.

[8] Countries are considered democratic if they are "full" or "flawed" democracies.

[9] Also considering low and lower-middle income countries adds four countries (e.g., the Philippines). As these countries were not democratic for most of the post-1950 period, it is appropriate to exclude them.

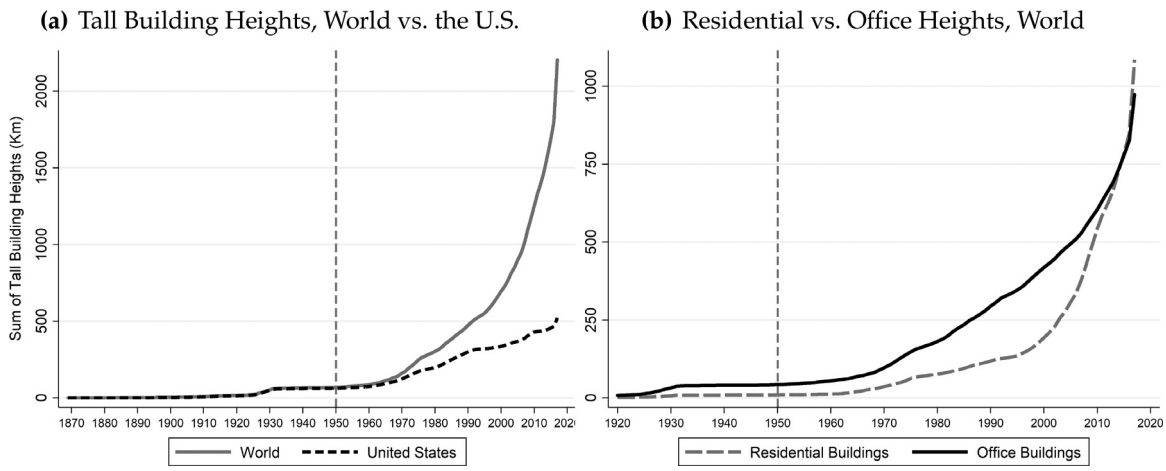


Fig. 1. TALL BUILDING HEIGHTS FOR THE WORLD, 1869-2017. *Notes:* SubFigure 1(a) shows the evolution of the stock of tall building heights (m) for both the world and the United States from 1869 to 2017. SubFigure 1(b) shows the world evolution of the stock of tall building heights (m) separately for residential and office buildings from 1920 to 2017. The dashed vertical line shows the year 1950, the start year of our main period of study (1950-2020). See text for details.

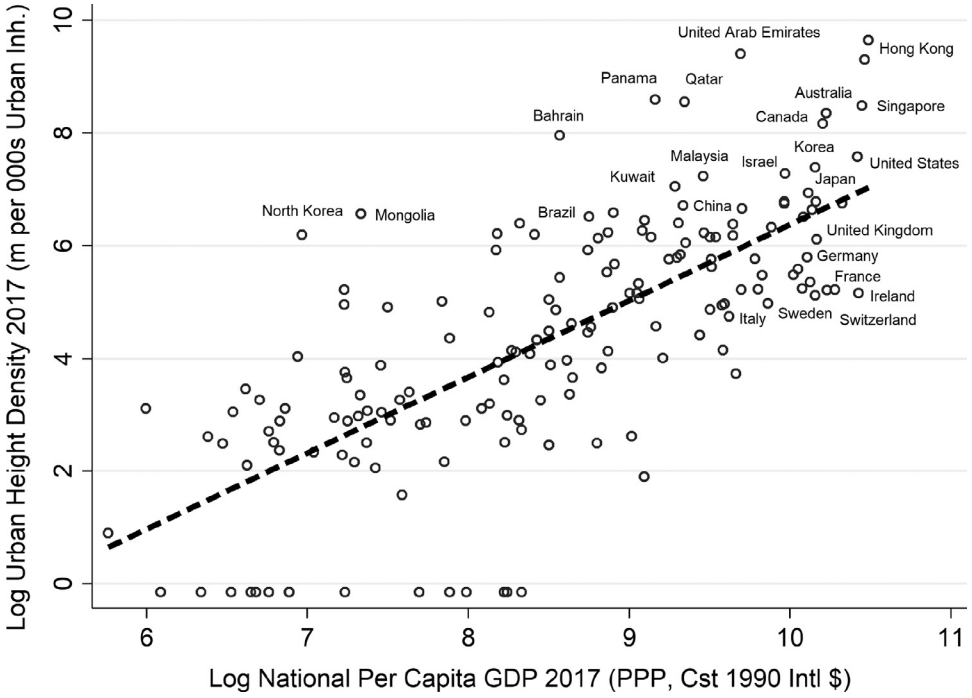


Fig. 2. URBAN HEIGHT DENSITY AND NATIONAL INCOME IN 2017. *Notes:* This figure shows the relationship between the log sum of tall building heights per urban capita (m per inh.) and log per capita GDP (PPP, constant 1990 international $) for 170 countries circa 2017.

we believe that this selection process is less arbitrary than us "cherry-picking" what we believe are more supply-elastic countries. Importantly, we will show below that other approaches lead to similar selections of countries and results.

Also, based on qualitative evidence, these choices make sense. Among high-income countries, Toronto now has "the third highest number of skyscrapers in North America" (CBC News, 2020). All Australian cities, not just Melbourne and Sydney but also Adelaide, Brisbane, Gold Coast and Perth, have a significant number of tall buildings (The Conversation, 2019). Seoul and other Korean cities such as Busan and Incheon have been at the forefront of skyscraper construction and innovation for some time now (NY Times, 2011; CNN, 2017). And Tel-Aviv's skyline is "undergoing dramatic transformation" (Globes, 2017). In middle-income countries, cities such as Panama City, Sao Paulo, Bangkok, and Kuala Lumpur, have undoubtedly some of the most impressive skylines in the world, especially given their country's middle-income status.

Lastly, while the U.S. and Japan are traditionally associated with skyscrapers, they have very high incomes and agricultural rents and large urban populations. They are above, but still close to, the regression line in Figure 2, thus suggesting that they do not have disproportionate stocks of tall buildings given their economic conditions. If anything, given their residual, they are only five and six countries away from being included in the benchmark set, respectively. The U.S. belonged to the benchmark set until 2010 and was among the top 10 countries in the world until 1980. As we will show below, one reason for the U.S. position is that California has experienced dramatic growth but still has relatively few tall buildings, thus offsetting the historical contributions of New York City and Chicago. Japan then also belonged to the benchmark set in 2010 (but not in 2020).

We do not include China because it is not democratic and some of its tall buildings may be considered as white elephants. While China is undeniably experiencing a construction boom (its residual improved

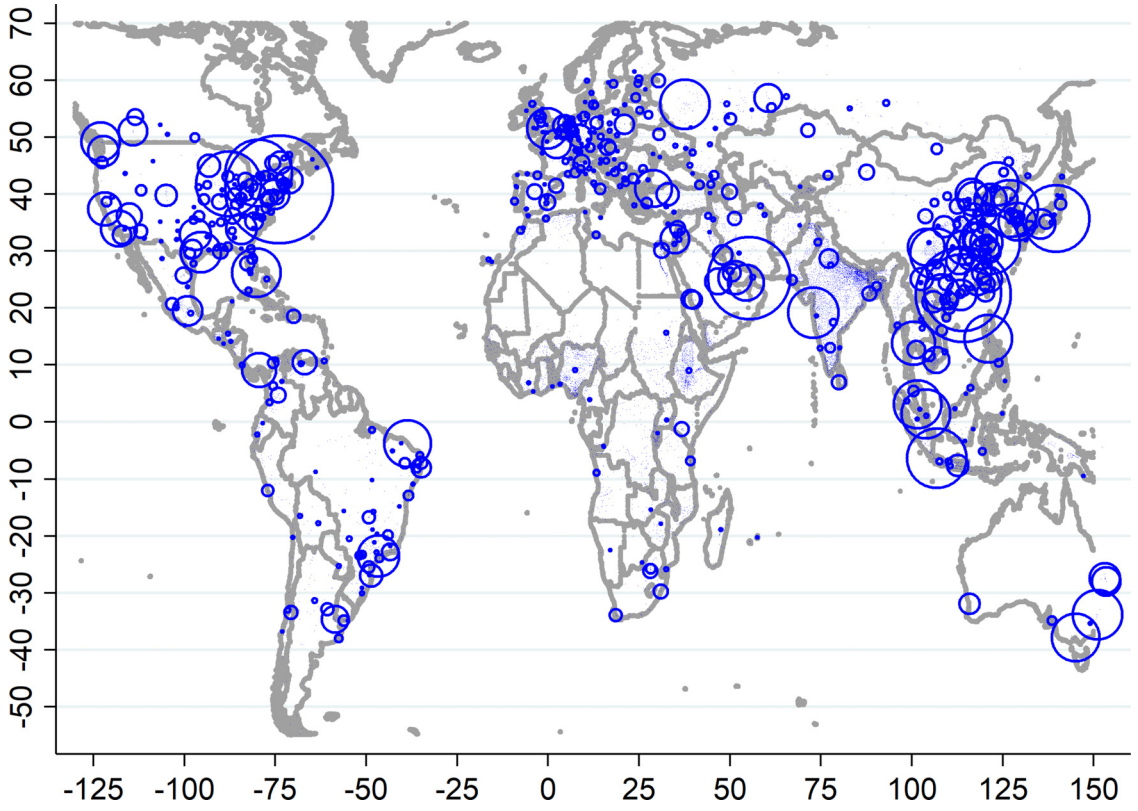


Fig. 3. Absolute Change in the Total Sum of Tall Building Heights (Km) 1950-2020. *Notes:* This figure shows for 11,719 agglomerations of at least 50,000 inh. in 2015 the absolute change in their total stock of tall building heights (km) between 1950 and 2020 (proxied by 2017; buildings above 80 meters only). Mean = 0.13 (km); Median = 0.00; SD = 1.91; Min = 0.00; Max = 145.41.

Table 1
Income and land rent and building heights, 1950-2020.

Dep. Var.:	Log Urban Height Density (m per 000s Urban Inh.) in Year t ($LUHTDENS_t$)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Countries	All	≥ 0	< 0	≥ p75 & DemUMH	≥ p75 & DemH	Residential ≥ p75		Office ≥ p75	
						DemUMH	DemH	DemUMH	DemH
$LUPCGDP_t$	0.49***	0.68***	0.37***	1.54**	3.23***	1.54***	2.66***	1.79**	1.61
	[0.10]	[0.10]	[0.16]	[0.67]	[0.61]	[0.34]	[0.47]	[0.55]	[1.32]
$LAGRENT_t$	0.13	0.30**	-0.06	0.55**	0.19	0.58**	0.40	0.47	0.28
	[0.09]	[0.13]	[0.10]	[0.26]	[0.40]	[0.22]	[0.42]	[0.31]	[0.61]
$LUHTDENS_{t-10}$	0.48***	0.46***	0.39***	0.46***	0.18	0.47***	0.45***	0.34*	0.40*
	[0.03]	[0.04]	[0.03]	[0.11]	[0.12]	[0.09]	[0.11]	[0.16]	[0.20]
Cntry FE, Yr FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	1,106	511	595	98	56	119	56	70	49
Countries	158	73	85	14	8	17	8	10	7
Adjusted R^2	0.79	0.80	0.76	0.87	0.91	0.80	0.86	0.84	0.83

Notes: Sample of 158 countries x 8 years (1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020) = 1,264 obs. Since we control for the dependent variable in t-10, we lose one round of data, hence N = 1,106 (col. (1)). "Dem" countries are "full democracies" or "flawed democracies" at any point in 2006-2017. "UM" and "H" countries are upper-middle income countries and high-income countries circa 2017, respectively. "0" and "p75"correspond to the following values of the selection residual: 0 and the 75th percentile. Col. (6)-(7) & (8)-(9): We study residential buildings and office buildings, respectively. Robust SEs clustered at the country level: * p<0.10, ** p<0.05, *** p<0.01.

dramatically since 1980), it has also experienced significant economic, agricultural and urban population growth. In 2020, it is only one country away from being above the 75th percentile value in the residual.[10] We will show below that results hold if we include the U.S., Japan, and China in the benchmark set.[11]

[10] While some cities overbuild relative to current populations (Wu et al., 2016; Glaeser et al., 2017), other cities like Beijing and Shanghai have limited their construction. On average, China is building at the rate required to absorb its growing urban population (Barr and Luo, 2021). Planning officials are relatively conservative in the allowable floor area ratios (FARs). Brueckner et al. (2017) find that the average land lease sales allow for residential and commercial FARs of about 2.4 on average. By way of comparison, Manhattan has a built FAR of 4.8, while New York City's built FAR is 1 (NYC Dept. of City Planning).

[11] We purposely do not include non-democratic countries in the benchmark set. Indeed, such countries may build more tall buildings for a given income level, making their height stock less responsive to economic conditions. In addition, they may build more tall buildings even in periods with slow or negative growth.

With the UMH sample (column (4)), the GDP coefficient is three times larger than in column (1), while the agricultural rent coefficient is four times larger. With the H sample (column (5)), the GDP coefficient doubles relative to column (4). The agricultural rent coefficient is insignificant due to its strong correlation with income at the country level. The coefficient of lagged urban height density is much lower than 1, indicating little persistence in tall building heights per urban capita in that sample. Lastly, the adjusted R^2 values increase from 0.79 in column (1) when we consider all countries to 0.91 in column (5) when we consider H countries only. This pattern shows the increasing explanatory power of the variables of interest when focusing on higher income countries.

To generate the gaps, we iterate each benchmark regression (column (4) or (5)) to get predicted heights for 2020, and then compare those heights to the actual 2020 data. The iteration proceeds as follows. Predicted log heights for 1960 are found by evaluating

$$\widehat{\text{LUHTDENS}}_{1960} = \alpha + \beta \text{LUPCGDP}_{1960} + \gamma \text{LAGRENT}_{1960} + \delta \text{LUHTDENS}_{1950}. \quad (1)$$

with LUHTDENS being log urban height density, LUPCGDP log per capita urban GDP and LAGRENT log agricultural land rent. The coefficients (α, β, γ, δ) are obtained from Table 1.

The year fixed effects are always included. However, for simplicity, they are omitted in writing (1). By construction, we ignore the country effects to compute the gaps. Other than that, (1) is exactly the same specification as we use for Table 1. To get predicted log building heights for 1970, we rewrite (1) with 1970 values for the first two covariates and with the 1960 predicted value playing the role of LUHTDENS_{1960}:

$$\widehat{\text{LUHTDENS}}_{1970} = \alpha + \beta \text{LUPCGDP}_{1970} + \gamma \text{LAGRENT}_{1970} + \delta \widehat{\text{LUHTDENS}}_{1960}. \quad (2)$$

The procedure continues until a LUHTDENS predicted value emerges for 2020. The building-height gap measure in 2020 is then equal to

$$\text{GAP}_{2020} = \widehat{\text{LUHTDENS}}_{2020} - \text{LUHTDENS}_{2020}, \quad (3)$$

or the difference in predicted and actual log height densities.

It is helpful to derive the connection between a gap and the change in the underlying building stock required to eliminate it. Letting Δ denote change, the answer is immediate from differentiating (3):

$$\Delta \text{GAP}_{2020} \approx -\frac{\Delta \text{UHTDENS}_{2020}}{\text{UHTDENS}_{2020}}, \quad (4)$$

which is denoted the *percentage-change gap*. If GAP = 2, then the change in the log of the existing urban height density required to eliminate the gap also equals 2. With (4) then equal to 2, it follows that unlogged urban height density must increase by 200% to close the gap, with 200 then denoted the percentage-change gap. The *per capita gap* (measured in km per urban inhabitant) is then found by using the corrected antilog of $\widehat{\text{LUHTDENS}}_{2020}$ to obtain $\widehat{\text{UHTDENS}}_{2020}$.

Finally, note that we compute the gaps for both the non-benchmark countries and the benchmark countries. By construction, the benchmark countries with a gap have small gaps (relative to the hypothetical "average" benchmark country).

4. Rankings

Table 2 ranks the top 20 countries in terms of the two gap measures just discussed and using the H set or the UMH set as our benchmark set. Columns (1) and (3) show the ranking based on the *percentage change gap* (the percentage change in height density required to close any gap). However, percentage changes are mechanically larger when the denominator is small. Thus, the percentage gap is mechanically larger in fast-growing countries with still relatively small height stocks today (e.g., Uzbekistan). If, instead, we focus on the ranking based on the absolute per urban capita gap (columns (2) and (4)), the list is now dominated by developed countries.

Various European countries are found in the list, which concords with common beliefs that they are more stringent in regulating heights than other nations (however, and, as we will examine below, other factors than land-use regulations could account for the gaps). Ireland, for example, has no buildings taller than 100 meters and only five buildings taller than 50 meters (Barr and Lyons, 2018), and this despite Dublin being one of the 10 wealthiest cities in the world and a financial hub (OECD, 2020). Paris has reluctantly embraced skyscrapers but has placed them outside of the city center in La Défense, in an attempt to keep them isolated from the old city, where skyscrapers like the Tour Montparnasse have proved controversial (Scicolone, 2012). Regulations in Switzerland, for example, give strong veto power to those opposed to skyscrapers, and, as a result, there are few in that country (Vogel-Misicka, 2011). For example, there are only five tall buildings in Zurich, again one of the wealthiest cities in the world and a financial hub (OECD, 2020). Many cities like London and Rome have implemented height caps so as not to block views of major monuments (Stewart, 2016). While these rules are being peeled back in some places, Europe remains slow to embrace tall buildings.

Indeed, as can be seen in Web Appx. Fig. A1, a few countries have seen their (logged) urban height density plateau out since the 1970s, including France, Germany, Ireland, Italy, Switzerland and, to a lesser extent, the United Kingdom, despite impressive economic growth (as shown by logged urban per capita GDP). Comparing it to Web Appx. Fig.A2 that shows the same patterns for the 14 UMH countries, one can see that their growth has not been less impressive than in the UMH set but that their urban height density has not increased fast enough to match their income growth. We believe that the *timing* of these results also makes sense. For example, in Paris the controversial Tour Montparnasse was completed in 1973. It became one of the most hated landmarks in the world and led two years later to the ban of new buildings over seven stories high in central Paris (NY Times, 2015), which can explain the plateau effect from 1980. Likewise, Dublin's Liberty Hall built in 1965 was immediately seen as one of Ireland's ugliest buildings (The Irish Times, 2019), which can explain the plateau effect from 1970.

In 2020, the data show 2,198 km of total height worldwide. Summing predicted values across countries, we get total predicted heights of 4,828 km, which generates a gap of 2,630 km – 6,000 Empire State Buildings (ESBs) – and a world gap factor of 2.2. However, negative gaps – i.e., an "excess" of heights given economic conditions between 1950 and 2020 – are observed in 45 countries where skyscrapers might be "white elephants" (e.g., the Gulf states, Mongolia, and North Korea, as can be seen circa 2017 in Figure 2). For 44 countries with a positive gap, the total gap is 3,143 km (7,250 ESBs). Next, if we use the UMH regression, we get a mechanically smaller predicted world total of 2,046 km. However, 31 countries still have a positive gap and their total gap is 919 km (2,100 ESBs).

While the percentage change is useful, we verify that the resulting ranking of countries is correlated with the ranking based on the absolute measure, i.e., the per capita gap in column (2) of Table 2. For the UMH and H benchmark sets, we obtain correlations of 0.65 and 0.66, respectively. However, if we weight the country gaps by the urban population of each country circa 2020, we obtain correlations of 0.77 between percentage and absolute rankings for both sets. Indeed, the urban population weights minimize the issue coming from low-stock countries having mechanically larger percentage gaps.

In the rest of the analysis, we use the percentage gap as our main measure. An issue with the absolute per urban capita gap is that it is generated by taking the antilog of the predicted values and then adjusting them by a correction factor. We follow the method of Wooldridge (2016, pp. 212–215). However, Wooldrige explains that this method is

We verify in Web Appx. Table A1 that the panel correlations with the economic variables disappear when restricting the analysis to "non-democratic" high-residual UMH or H countries.

Table 2
Countries with the largest umh- and h-based gaps 2020.

	(1) Percentage Change		(2) Per Capita Gap_H		(3) Percentage Change		(4) Per Capita Gap_{UMH}	
	Gap_H		(Km per Mil. Urb. Inh.)		Gap_{UMH}		(Km per Mil. Urb. Inh.)	
Rank	Country	Gap	Country	Gap	Country	Gap	Country	Gap
1	Ireland	4.88	Ireland	21	Mauritius	6.06	Mauritius	16.1
2	Mauritius	4.51	Mauritius	20	Uzbekistan	5.14	Taiwan	12.4
3	Slovenia	3.69	Austria	12	Taiwan	4.57	Netherlands	4.9
4	Switzerland	3.61	Taiwan	12	Switzerland	4.15	Sri Lanka	3.4
5	Uzbekistan	3.37	Sri Lanka	8	Ireland	4.13	S. Korea	3.4
6	Norway	3.21	Trinidad	6	Slovenia	4.06	Ireland	1.6
7	Austria	2.9	Switzerland	6	Italy	3.93	Switzerland	1.5
8	Taiwan	2.78	United States	6	Netherlands	3.72	Austria	1.4
9	Sweden	2.77	Slovenia	5	Sri Lanka	3.58	Italy	1.3
10	Sri Lanka	2.61	Norway	4	P. Rico	3.58	Slovenia	1.2
11	Italy	2.53	South Korea	4	Syria	3.5	Japan	1.1
12	Denmark	2.52	United Kingdom	3	Guatemala	3.47	France	0.8
13	Trinidad	2.5	Netherlands	3	Armenia	3.37	Belgium	0.6
14	France	2.49	Estonia	3	France	3.26	United Kingdom	0.6
15	Germany	2.47	Germany	3	Denmark	3.21	Denmark	0.6
16	Eq. Guinea	2.43	Sweden	3	Lesotho	3.18	Germany	0.6
17	Finland	2.23	France	2	Portugal	3.06	Armenia	0.5
18	United Kingdom	2.15	Denmark	2	India	3.05	Trinidad	0.4
19	Lesotho	2.12	Italy	2	Germany	2.98	Portugal	0.3
20	Portugal	2.02	Slovakia	2	S. Korea	2.96	India	0.2

Notes: Col. (1) & (3): The gap is the percentage change in urban height density required to make the height stock similar to the selected benchmark set of countries (e.g., 4.88 means 488%). Col. (2) & (4): The gap is expressed in km of heights per million urban capita.

imperfect and rests on many assumptions. Notwithstanding, the high correlation between the percentage and absolute gaps suggests that the results that we establish below should not depend much on whether we use one or the other. In addition, in the rest of the analysis, we will also often use urban population weights.

Finally, and importantly, the ranking of countries does not depend much on whether we use UMH or H. Indeed, for the same measure, the correlation between the UMH-based ranking and the H-based ranking is 0.85-0.89.

City-Level Analysis. Global historical data on city incomes and agricultural land rents at the edge of cities do not exist, ruling out a worldwide, within-country study. However, our benchmark set of countries may not be sufficiently representative of the rest of the world in terms of city size distribution, which matters because income and land rent should affect tall building construction more in larger cities. However, if we use the *Global Human Settlement* (GHS) data set of 11,719 urban agglomerations for the year 2015, Kolmogorov-Smirnov tests suggest that the Kernel distribution of city population sizes is not significantly different between the benchmark and non-benchmark countries for both sets (not shown). Next, we can directly use the GHS data set of 11,719 cities x 4 years with city population data (1975, 1990, 2000, 2015) to perform a global analysis at the city level.

Focusing on the year 2015, we first regress the log of city building heights (km) on six dummies showing if the city has 55-100K, 100-500K, 500-1,000K, 1,000-5,000K, 5,000-10,000K or 10,000K+ inhabitants (50-55K is the omitted category), while including country fixed effects, with the results illustrated diagrammatically. As seen in Fig. 4, the relationship is non-linear, with heights significantly increasing after 100K. The figure also shows the same relationship for total heights in the central city (e.g., New York City for the New York-Newark-Jersey City metro area) vs. peripheral areas (e.g., Newark and Jersey City). The overall relationship is, as expected, driven by central areas.

To define the set of "benchmark" cities, we identify cities that have disproportionately more heights for a given population size and level of national urban income and land rent. For the year 2015, we regress the log sum of heights on the six dummies as well as log urban per capita GDP (PPP, constant international $) and log agricultural land rent (both defined at the country level), which we also interact with the six dummies.[12]

Doing so, we obtain 302 cities belonging to a high-income democratic country and whose residual is above the 75th percentile value. Including upper-middle income countries, we get 384 cities. We find a strong correlation between the country- and city-level residual values (0.68). However, the correlation is below 1 because cities have different values within a same country. For example and as expected, New York City and Milan are included in the benchmark sets whereas Los Angeles and Rome are not. One advantage of this analysis is that it includes in the benchmark set cities that were not included at the country level. One disadvantage is that it ignores the possibly low substitutability in building space demand across cities within countries, which country-level regressions do. For example, even if building up is easier in Milan, it may not have a large impact on Italy's total gap. Most Italian cities have disproportionately few tall buildings for their economic conditions, and whether Italy's gap is reduced by Milan's higher supply elasticity ultimately depends on demand substitutability across cities.

For the H and UMH sets, we then use panel regressions 1975-2015 where the dependent variable is the log sum of heights in year t and the variables of interest are the urban income and land rent variables in year t, which we interact with three dummies for whether the city has 100-500K, 500-1,000K and 1,000K+ inhabitants in 1975, respectively. We use three dummies to reduce the number of coefficients we have to track. We use 1975 because post-1975 changes in city populations are endogenous to post-1975 changes in the country gaps. We then include a lag of the log sum of heights as well as city and year fixed effects and cluster standard errors at the country level.[13]

As expected, the coefficient of income is higher for larger cities. For the H set, we find 0.16 for the omitted category (50-55K) and 1.28*, 2.26*** and 2.63*** for the 55-500K, 500-1,000K and 1,000K+ cate-

[12] For agglomerations a belonging to population categories p and for the year 2015, the model is: $\text{LHEIGHT}_a = \alpha + \Sigma_p \beta_p \mathbb{1}(\text{POP}_a = p) \times \text{LUPCGDP}_c + \Sigma_p \gamma_p \mathbb{1}(\text{POP}_a = p) \times \text{LAGRENT}_c + \mu_a$.

[13] For agglomerations a, population categories p and years t, the model is: $\text{LHEIGHT}_{a,t} = \alpha + \Sigma_p \beta_p \mathbb{1}(\text{POP}_{a,1975} = p) \times \text{LUPCGDP}_{c,t} + \Sigma_p \gamma_p \mathbb{1}(\text{POP}_{a,1975} = p) \times \text{LAGRENT}_{c,t} + \text{L1.LHEIGHT}_{a,t} + \theta_a + \kappa_t + \mu_{a,t}$.

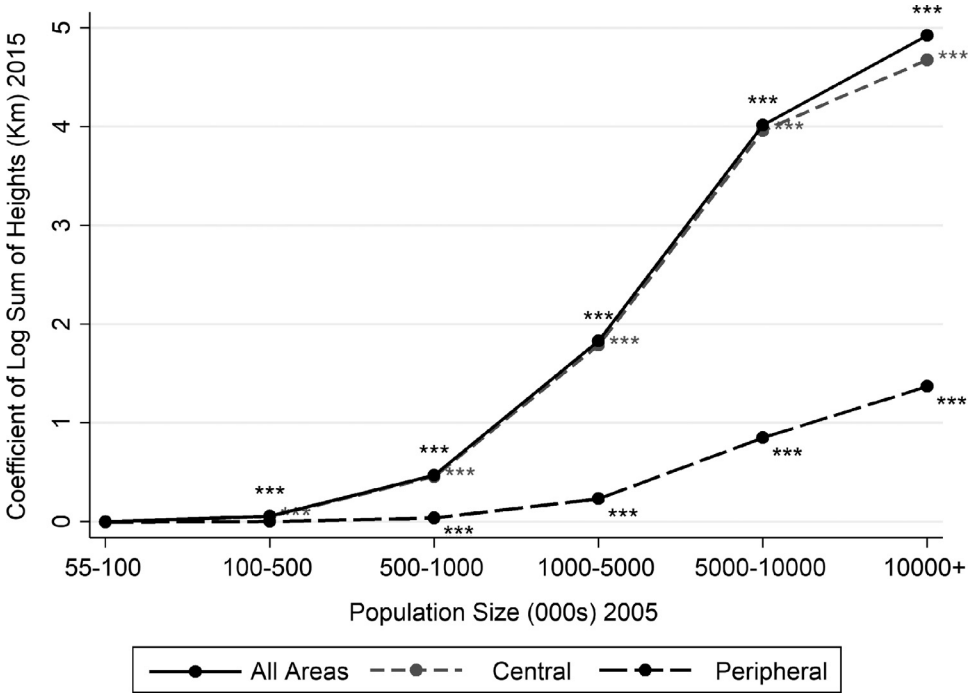


Fig. 4. CITY BUILDING HEIGHTS-POPULATION RELATIONSHIP, 2015. *Notes:* This figure shows for 11,719 agglomerations of at least 50,000 inhabitants in 2015 the relationship between the log sum of tall building heights (km) and the population size category (the omitted category is 50,000-55,000) ca. 2015. The 11,719 agglomerations in 2015 belong to 158 countries.

gories, respectively. We find a similar gradient for the UMH set (-0.39, 0.85, 1.81** and 2.06***). Given the total population shares of the different categories of cities, the average coefficient is 2.32*** and 1.81***, respectively. While it is lower than in Table 1, it remains high. Indeed, 1,000K+ cities account for almost three fourths of urban populations. We then do not observe any gradient for land rent, and the average coefficient is -0.60 and -0.12, respectively. The land rent coefficient is now negative, suggesting that the national land rent control may be really imperfect for city-level analyses. Lastly, the coefficient of past heights is about 0.20 (**) in both cases.

Using the data and the estimated coefficients, we can proceed as we did in the previous section to obtain the percentage-change gap of each of the 11,719 world cities in each year. Using their population size as weights, we can then obtain the average gaps for each country in each year and compare them to the ones we obtained with the country-level analysis. We find a correlation of about 0.6 and 0.4 for the H- and UMH-based gaps. For the H-based gaps, the correlation is relatively high. Yet, it is lower than one. However, the discrepancy could be due to: (i) the different period (1975, 1990, 2000 and 2015 vs. every ten years in 1950-2020); (ii) the benchmark sets being different; and/or (iii) the fact that the city-level analysis ignores the possibility of low within-country demand substitutability.

Nonetheless, the high correlation for the H-based gaps gives us confidence in the gaps used so far. The city-level analysis also highlights how larger cities drive the gaps. For example, Ireland has the highest gap for both the country- and city-level analyses. The gap is then five times higher for Dublin than for Cork. In Italy, among million-plus cities, Rome has a higher gap than Turin, and Milan has the lowest gap but its gap is still high.

Finally, the country-level analysis may over-estimate the gaps in large countries because their cities have more area to grow (hence not necessitating tall buildings as much) and because land rent at the edge of cities is mismeasured for them. Studying whether the difference between the country- and city-based country gaps increases with log land area or the log number of cities (N = 158), we find negative and almost nil coefficients (-0.31/-0.26, not significant; $R^2 \leq 0.02$). Overall, since the city-level analysis also has limitations, we focus on the country-level analysis in the rest of the study.

Office vs. Residential Towers. Columns (6)-(7) of Table 1 replicate columns (4)-(5) of Table 1, but with the height variable computed using residential buildings, while columns (8)-(9) use office buildings (the benchmark countries are selected using residuals that are specific to each building type). For the UMH set (cols. (6) and (8)), point estimates are similar for the two types of buildings. For the H set (cols. (7) and (9)), the coefficient of income is higher for residential buildings. When using these estimates to compute the percentage-change gaps, we find that the H-based gaps are overall higher for office buildings than for residential buildings (Figure 5(a) shows their kernel distribution in 2020). However, only considering positive gaps, no significant difference is obtained. The lower residential gaps observed globally are thus explained by negative gaps (i.e., excesses) in various countries. For example, Middle-Eastern and Latin American countries have disproportionately large residential tall-building stocks, possibly as a result of "white elephant" projects and/or preferences for high floor luxury condos. Finally, countries with higher, or lower, residential gaps tend to have higher, or lower, office gaps (correlation of 0.70-0.80 depending on the benchmark set used).

Richer vs. Poorer Countries. Cols. (1) and (4) of Table 3 show higher gaps in richer countries today (2020), i.e., countries with a higher log total per capita GDP (PPP, constant 1990 international $). Cols. (2)-(3) and (5)-(6) show that the residential gaps are more correlated than the office gaps with economic development.[14] Alternatively, restricting the sample to developed countries, we find that residential gaps tend to dominate office gaps (see Figure 5(b)). Thus, developed country cities might be more open to creating jobs than receiving new residents (with the caveat that this assumes that the gaps also capture regulatory constraints). One possibility is that in more developed countries, while both businesses and residents pay local taxes, local governments view residents as "costlier," since many services (e.g., schooling, sanitation, water, etc.) are publicly provided.

[14] For countries c, the model is: $GAP_{c,2020} = \alpha + \beta\ LPCGDP_{2020} + \mu_c$. Urban population weights minimize the low-stock issue and make the analysis representative of the global distribution of urban populations.

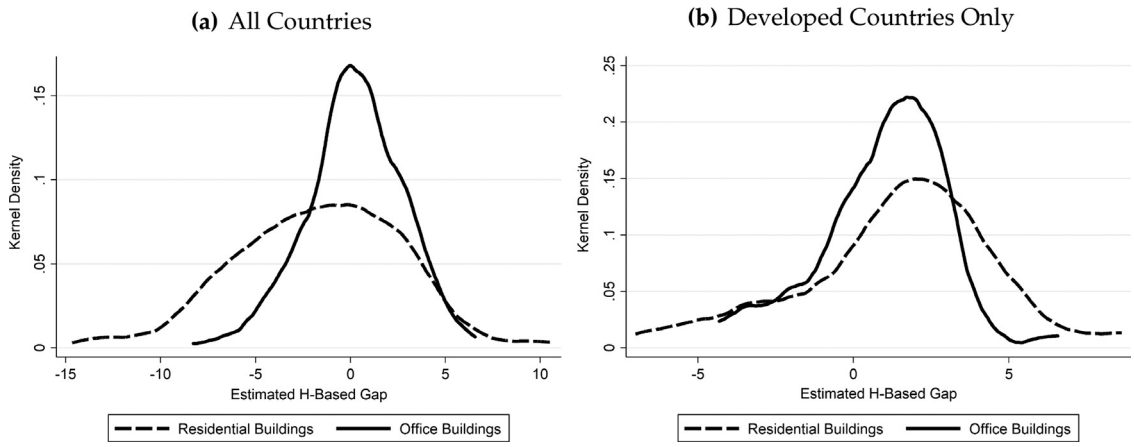


Fig. 5. GLOBAL DISTRIBUTION OF THE RESIDENTIAL AND OFFICE GAPS, 2020. *Notes:* The left panel shows for the 158 countries of our sample circa 2020 the Kernel distributions of the H-based percentage-change gaps separately for residential buildings and office buildings. The right panel shows the same Kernel distributions for 47 developed (i.e., high-income) countries as of 2020.

Table 3
The gaps and economic development, 2020.

Dependent Variable:	H-Based Pct-Change Gap 2020			UMH-Based Pct-Change Gap 2020		
Considered Gap:	(1) All	(2) Resid.	(3) Office	(4) All	(5) Resid.	(6) Office
LPCGDP 2020	1.88***	2.48***	0.62**	0.97***	0.92***	0.69**
	[0.20]	[0.27]	[0.26]	[0.32]	[0.33]	[0.32]
Observations	158	158	158	158	158	158
R^2	0.58	0.55	0.10	0.16	0.12	0.09

Notes: LPCGDP 2020 is log national per capita GDP (PPP, cst 1990 intl $) in 2020. Columns (2)-(3) and (5)-(6): We consider the gaps based on residential buildings (Resid.) or office buildings (Office) only. We use urban population weights (2020). Robust SEs.

The U.S. Case. These patterns could explain the U.S. gap. Many tall buildings were built before 1950 and subsequent U.S. construction may have failed to match income growth to the same extent as in the "best" countries. As a result, with the gaps that we compute thereby influenced, it may not be surprising that the U.S. gaps are large. To assess the validity of the methodology, we perform our panel analysis for 50 states (1930-2020; decadally). The U.S. gap is driven by California. If California were like the "best" U.S. states, the U.S. would be ranked 20th in per urban capita H-based gap instead of 8th.[15]

We now turn to generating results on the determinants and effects of the gaps. Finally, we assess how sensitive our results are by implementing various robustness checks.

5. What Constraints May the Gaps Capture?

The gaps could be due to differences in construction costs, geographical constraints, dispersed ownership, land-use regulations, and institutional and cultural factors more broadly. We examine how the gaps correlate with variables proxying for these dimensions. More precisely, we regress the gaps in 2020 on each variable. We first consider each variable one at a time as there is often limited data for each dimension. For selected variables, we then consider some of the variables simultaneously. Note that we always control for log total per capita GDP in 2020 (PPP, constant international 1990 $), as the goal is to understand why some countries have higher gaps for a given level of economic development.[16] We then use urban population weights (2020) because it is important to study what drives the world gap, hence making the analysis representative of the global distribution of urban populations. Lastly, one caveat is that the gaps are generated using the coefficients of Table 1, which are by construction imprecisely estimated given that we only select a few benchmark countries. As such, the standard errors in the rest of the analysis are too small, and the results should be taken with caution.

The results are reported in Table 4 with each panel corresponding to a separate factor. For each panel, the first column reports the coefficient when using the H-based gaps while the second column reports the coefficient when using the UMH-based gaps.

Possible Factors. First, conditional on income, we do not find that the gaps are explained by differences in construction costs. World Bank (2019a), which is responsible for producing PPPs globally, reports the price level of broad consumption categories in 2011 (relative to the world = 100). In Panel A we consider the price level of construction, which includes information on materials, labor, and equipment. As seen, the coefficients are negative, not positive, and close to nil. Another important aspect of construction costs is financing. In Panel B, we consider the countries' average lending rate in 1990-2020 (International Monetary Fund (2021); data often unavailable before).[17] In Panel C, we alternatively consider the 1980-2018 average of the fi-

[15] For more details on the U.S. analysis, see Web Appx. Section C. If we use the same specification as in Table 1 but for the U.S. only (N = 8), the coefficient of urban income is 50% smaller than for UMH (0.99*** vs. 1.54**) whereas the coefficient of land rent is similar (0.68* vs. 0.55**). If we study the U.S. between 1870 and 1940 (N = 7; excl. the World War II decade), we obtain higher coefficients, at 2.74** and 2.22**. Skyscrapers in the U.S. were much more responsive to economic conditions before 1950. In fact, many land use regulations were adopted in the 1960s (e.g., New York City reformed its zoning ordinance in 1961).

[16] For countries c and factor X, the exact model is: $GAP_{c,2020} = c + \phi_X X_c + \kappa LPCGDP_{c,2020} + \mu_c$.

[17] It "is the rate charged by banks on loans to the private sector." Due to high inflation in some countries, we remove outlying observations with a lending rate above the 90th percentile value in the sample.

Table 4
Possible determinants of the gaps, 2020.

Dependent Variable: Pct Change-Change Gap 2020 (1st Column = H-Based; 2nd Column = UMH-Based)										
RHS:	A Log Const Cost		B Log Lend Rate		C Fin Inst Dvt		D Log Earthquake		E Log Bedrock	
β	-0.00	-0.01	-0.09	-0.52	-0.98	-1.58	0.38	0.77	-0.63	-0.77
	[0.00]	[0.01]	[0.40]	[0.55]	[1.40]	[2.27]	[0.49]	[0.64]	[0.46]	[0.64]
Obs.	146	146	114	114	147	147	156	156	158	158
RHS:	F Share Mountain		G Log Foundation		H Pop Sh. 1800		I Pop Sh. 1950		J WW2 Sh. Bomb	
β	-0.00	0.00	0.92***	1.42***	0.11*	0.15*	-0.01	-0.00	0.32	1.41**
	[0.02]	[0.02]	[0.25]	[0.29]	[0.06]	[0.08]	[0.00]	[0.01]	[0.49]	[0.68]
Obs.	158	158	158	158	158	158	158	158	158	158
RHS:	K WW2 Mort Rate		L Log Cult WHS		M Log Mix WHS		N Log Nat WHS		O Log Tourists	
β	-0.01	-0.02	0.51**	0.64*	-0.69*	-1.01*	-1.08***	-1.84***	-0.28	-0.37
	[0.06]	[0.10]	[0.25]	[0.38]	[0.39]	[0.54]	[0.28]	[0.38]	[0.24]	[0.33]
Obs.	158	158	157	157	157	157	157	157	152	152
RHS:	P Max FAR		Q Quality Ctrl		R Hous P. Ela.		S Speed Hous S.		T Democracy	
β	-0.13***	-0.20**	0.02	0.02	-0.60	-1.00	-1.05	-2.54*	0.12	0.20
	[0.01]	[0.02]	[0.02]	[0.02]	[0.39]	[0.67]	[0.99]	[1.28]	[0.62]	[0.88]
Obs.	49	49	155	155	21	21	21	21	158	158
RHS:	U Polity Score		V Rule of Law		W Ctrl Corrupt.		X Log Urb Plan.		Y Home Own.	
β	0.03	0.02	0.29	0.19	-0.07	-0.34	0.22	0.02	0.01	0.01
	[0.05]	[0.07]	[0.36]	[0.56]	[0.24]	[0.42]	[0.22]	[0.32]	[0.01]	[0.02]
Obs.	156	156	157	157	157	157	158	158	105	105

Notes: This table shows the correlations between the H-based gaps (1st column of each panel) or the UMH-based gaps (2nd column) and various potential determinants of the gaps. We always control for log national per capita GDP (PPP, cst 1990 intl $) ca. 2020 and use urban population weights. Each coefficient corresponds to a separate regression. Robust SEs: * p<0.10, ** p<0.05, *** p<0.01.

nancial institutions index of the International Monetary Fund (2021). As seen, no significant difference is observed.

Next, we examine if, conditional on income, geographical constraints could explain the gaps. Pagani et al. (2018) provides at a fine spatial resolution global maps of earthquake risk.[18] Using the *Global Human Settlement* (GHS) database of European Commission (2018), we obtain the GIS boundaries and 2015 population of all (11,719) 50K+ agglomerations today. Finally, we use these data to estimate population-weighted averages of earthquake risk for each country. Such measures do not correlate with the gaps (Panel D). Likewise, Shangguan et al. (2017) provides at a fine spatial resolution global maps of bedrock depth. Using the GHS data, we construct population-weighted averages of bedrock depth for each country. No significant difference is observed (Panel E). Lastly, GRID-Arendal (2016) reports the population-weighted percentage of mountain area per country. Ruggedness could negatively affect buildability or increase the need to build taller due to the lack of flat areas. We find no correlation (Panel F).

Indeed, historically important geographical constraints were, thanks to technological progress, largely overcome by 1960. During earthquakes, structural techniques allow buildings to sway without damaging the structure (Wang, 2016). While bedrock depth may influence skyscraper locations locally (Rosenthal and Strange, 2008; Barr et al., 2011), it is unlikely to systematically impact building construction at the country level. Indeed, more than half of the variation in bedrock depth at the GHS agglomeration level comes from *within* countries. Thus, while some locations might have overly deep bedrocks, that should not be the case of *all* locations in a same country. Furthermore, piles have been used to anchor skyscrapers for over a century (Bradford and Landau, 1996). While being below sea level makes foundation work costlier if soils are porous (Ibid.), this condition only concerns 0.2% of global urban land. Finally, foundation costs are a relatively small part of the overall construction and land costs in central city neighborhoods.

Panel G considers the log of the number of years since the country's largest city was founded.[19] As seen, the gaps are strongly correlated with historical settlement. In Panel H (I), and focusing again on each country's largest city, we use the ratio of their 1800 (1950) population and their 2020 population (x 100). A higher ratio indicates that more of the city already existed by 1800 (1950) (obtained using the historical population data from Jedwab et al. (2021)).[20] As seen, we find some effects for 1800 and no effects for 1950, suggesting that countries with older (mostly pre-industrial) cities have larger gaps.

Why do countries with more historical cities have higher gaps? It could be due to historically dispersed ownership constraining land assembly and redevelopment or land-use regulations aimed at protecting historical areas. To our knowledge, there are no globally available measures of dispersed ownership. However, we can test if countries that experienced relatively higher destruction rates during World War II (WW2) have lower gaps today. Since no such measures exist, we obtain from Wikipedia a list of bombings that took place during WW2 and their death toll, and then compute bombing-related mortality rates at the country level (as % of 1939 population). As seen in Panel J, we do not find that the gaps are negatively correlated with WW2 bombings. Since the Wikipedia list may be incomplete, in Panel K we try the overall mortality rate (Ibid.), finding no correlation. However, the lack of data on dispersed ownership in "normal" (non-WW2) times implies that we cannot discard that specific mechanism.

[18] "Global Earthquake Model Global Seismic Hazard Map (version 2018.1) depicts the geographic distribution of the Peak Ground Acceleration with a 10% probability of being exceeded in 50 years."

[19] We obtain this information by consulting the Wikipedia webpage of each city.

[20] Jedwab et al. (2021) rely on data from Chandler (1987) and Wikipedia (2020d).

We then test if the gaps are higher in countries with valuable historical areas in their cities. In Panels L, M and N, we consider the log number of cultural, mixed and natural *World Heritage Sites* (WHS) ca. 2020 (UNESCO, 2020), while controlling for log total population in 2020 (mixed WHS "contain elements of both natural and cultural significance"). As seen, the gaps are positively correlated with the number of cultural WHS, which include the "historic centres" of Paris, Rome, and Vienna. The gaps are then strongly negatively correlated with natural WHS. This might be because countries with natural WHS implement policies that enforce the protection of natural areas and prevent sprawl, which should in turn lead to taller buildings, hence lower gaps.

Historical areas' value may be due to the tourism revenues they generate. However, in Panel O we do not find that the gaps correlate with the log of the average number of tourist arrivals during the period 1990-2020 (World Bank (2019b); data unavailable before), while controlling for log total population in 2020. We obtain similar non-results if we use tourism GDP data for the same period (World Bank, 2019b) (not shown), which is not surprising given that correlation of the log of tourist arrivals and the log of tourism receipts is 0.9. Indeed, while cultural WHS may be associated with higher gaps, non-cultural WHS associated with lower gaps also bring in tourists. Tall buildings, hence lower gaps, may also generate tourism, as exemplified by New York City or Hong Kong.

Thus, it is likely that historically more urbanized countries adopt land-use regulations to protect their valuable historical areas. Countries could also constrain tall building construction for other reasons, for example to create more open cities in which people can better see the sky. To our knowledge, there are almost no direct international measures of land use regulations. Nonetheless, in Panels P-S we present regression results where we regress the gaps on several indirect land-use regulation variables.

We first use a regulatory database established by Shlomo Angel.[21] Angel's data set contains 195 cities worldwide with at least 100,000 population in 2015. The database includes maximum Floor Area Ratio (FAR) (N = 95), maximum building height (N = 114), and the maximum number of dwellings per acre (N = 35). However, one limitation of this data set is that information is available only for the peripheral areas of cities. Therefore, to be able to use these variables, we must assume that they can serve as good proxies for the same variables in the central areas of the cities (e.g., cities with relatively low FARs in their peripheral areas have relatively low FARs in their central areas). The data set is not ideal but there is to our knowledge no other global database of land-use policies.

Combining the information from these variables gives information on building-height regulations for 138 cities in 51 countries, using the maximum FAR as the main measure. For cities for which we know maximum building height but not the maximum FAR, we predict the maximum FAR from a simple regression. We also use such a prediction based on maximum number of dwellings to gain more cities for the sample. For countries with multiple cities in the data, we average the maximum FAR values using the population of each city ca. 2015, which yields country values for 49 out of our 158 countries.

As can be seen in Panel P, our gaps show the expected negative relationship to this regulation measure, with a higher maximum FAR associated with a lower building-height gap. Next, we consider a measure taken from the *Doing Business* website that capture the "procedures, time and cost to build a warehouse," which constitute the only measures of land-use regulations that could be obtained from World Bank data. Panel Q shows that the H-based gaps are higher if building regulations are of higher "quality," hence more stringent. However, the correlations are not significant. Lastly, from Caldera and Johansson (2013) and for 21 OECD countries only, we obtain measures of the elasticity of the price responsiveness of housing supply (Panel R) and of the speed of housing supply adjustment (Panel S). We find negative correlations, suggesting lower gaps when housing markets are more responsive. The correlations are mostly insignificant given N = 21.

As discussed in Section 4, some countries may have negative gaps due to "white elephants". However, conditional on income, we do not find that the gaps correlate with: (i) a dummy if the country was democratic at any point in 2006-18 (The Economist, 2018) (Panel T); (ii) the average polity score of the country in 1950-2018 (Center for Systemic Peace, 2021) (Panel U); and (iii) the average rule of law index (Panel V) or control of corruption index (Panel W) of the country in 1996-2019 (World Bank, 2021). Thus, institutions specific to land use may matter, not institutions in general.[22]

Simultaneous Inclusion. In Table 5, we examine how the estimated coefficients of selected factors vary as we include them simultaneously. We consider the log number of years since a country's largest city was founded, the log number of cultural WHS, and the maximum FAR, while controlling for log per capita GDP and log total population. As can be seen in cols. (1)-(3) and (6)-(8), they all significantly correlate with the gaps when included one by one. When simultaneously included, sample size obviously decreases. As can be seen in cols. (4) and (9), the correlation with cultural WHS turns negative due to possible collinearity when also including the years since foundation variable. We thus omit the cultural WHS variable in cols. (5) and (10). The coefficient of the FAR variable then remains unchanged throughout. Overall, the results suggest that countries with older cities have higher gaps, either due to land-use regulations aimed at protecting historical areas and/or dispersed ownership. Land-use regulations are also independently correlated with the gaps, indicating that high-gap countries might also have more stringent land-use regulations unrelated to protecting historical areas.

Land-Use Regulations. We now use our data to generate additional results related to land-use regulations. First, could historical reliance on urban planning be a possible explanation for the apparently important role of land-use regulations? Urban planning became popular in the 19th century as a response to uncontrolled urbanization. While some urban planners follow Le Corbusier in thinking that tall buildings are needed in order to make cities more compact, many "low-rise" urban planners follow Jane Jacobs (Jacobs, 1961) in seeing tall buildings as a threat to neighborhood quality (Glaeser, 2011). In Panel X of Table 4, we thus consider the logged number of renowned urban planners (Wikipedia, 2020e) while controlling for log urban population. While we find a positive correlation for the H-based gaps, it is not significant. The correlations only slightly increase, and remain insignificant, if we assign to formerly colonized developing countries the logged number of planners of their colonizer (not shown), to account for the fact that their land-use regulations may date from the colonial period.

Second, the so-called homevoter hypothesis says that homeowners, worried that increases in supply reduce property values, lobby local governments to impose regulations (Fischel, 2001). To explore this idea as a source of the gaps, in Panel Y we consider the home ownership share today (sources: Wikipedia (2020a,c); HOFINET (2020); available for 105 countries). As seen, no correlation is found, consistent with Fischel's conjecture that his hypothesis mostly concerns local communities and likely does not apply to larger cities and urban systems as a whole.

Third, Angel's database provides information on other land use regulations than the maximum FAR. More precisely, we know for the cities of 49 countries if: (i) there is a greenbelt (GB) or an urban growth boundary (UGB); (ii) there are strong zoning laws; (iii) if the government acquires land to plan for urban land expansion; and (iv) if there is a minimum allowable plot size for construction. As before, we obtain country values

[21] Note that the results of Angel's survey have not been published yet. We obtained the data by contacting Shlomo Angel (https://marroninstitute.nyu.edu/people/shlomo-solly-angel) directly.

[22] We find similar non-results with other measures, including the World Bank Governance Indicators, the Opacity Index of PricewaterhouseCoopers/Milken, the Global Opportunity Index of the Milken Institute, and the JLL Real Estate Transparency Index (Web Appx. Table A2).

Table 5
Determinants of the gaps, simultaneous inclusion, 2020.

Dep. Var.:	H-Based Pct Change-Change Gap 2020					UMH-Based Pct Change-Change Gap 2020				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Log Yrs Founded	0.84***			1.23**	0.90***	1.29***			2.05***	1.46***
	[0.23]			[0.53]	[0.31]	[0.26]			[0.59]	[0.37]
Log Cultural WHS		0.51**		-0.59			0.64*		-1.05	
		[0.25]		[0.55]			[0.38]		[0.68]	
Max FAR			-0.12***	-0.11***	-0.14***			-0.19***	-0.17**	-0.21***
			[0.04]	[0.04]	[0.04]			[0.06]	[0.07]	[0.06]
Obs.	157	157	49	49	49	157	157	49	49	49
R^2	0.69	0.63	0.58	0.68	0.68	0.46	0.29	0.29	0.56	0.53

Notes: This table shows the correlations between the H- or UMH-based gaps and various potential determinants of the gaps. We always control for log national per capita GDP (PPP, cst 1990 intl $) and log total pop. ca. 2020 and use urban pop. weights. Robust SEs.

Table 6
Gaps and land use regulations, 2020.

Dependent Variable:	H-Based Gaps		UMH-Based Gaps		Corr w / Max FAR
	(1)	(2)	(3)	(4)	(5)
Maximum Floor Area Ratio (FAR)	-0.13***	-0.19***	-0.20**	-0.27**	
	[0.006]	[0.003]	[0.016]	[0.014]	
Dummy Green Belt (GB)		-0.67		-0.83	-0.15 n.s.
		[0.448]		[0.527]	
Dummy Urban Growth Boundary (UGB)		-1.46*		-1.38	-0.12 n.s.
		[0.053]		[0.208]	
Dummy Full or Partial Zoning		0.63		1.04	-0.03 n.s.
		[0.376]		[0.405]	
Dummy Gvt Land Acquisition		0.68		0.83	-0.09 n.s.
		[0.272]		[0.421]	
Dummy Min Plot Size Regulation		-2.02**		-2.43*	-0.20 n.s.
		[0.027]		[0.085]	
Observations	49	49	49	49	49
R^2	0.51	0.66	0.12	0.28	_

Notes: Columns (1)-(2) and columns (3)-(4) show the conditional correlation between the H-based gaps and the UMH-based gaps and the maximum FAR values and the values of the other land-use policy variables. Column (5) shows the individual conditional correlations between the maximum FAR values and the values of the other land-use policy variables. We control for log national per capita GDP in 2020 and use urban pop. weights (2020). Robust SEs: n.s. (not significant) p>0.10, * p<0.10, ** p<0.05, *** p<0.01.

by averaging the values of these variables using the population of each city circa 2015.

Using these data, we answer two questions. First, which land-use regulations do the gaps capture? Second, do countries with stringent height restrictions "compensate" their urban residents by having lenient regulations in other dimensions?

First, Table 6 show how the gaps correlate with the max FAR variable and the other land-use policy variables. If we include all variables (cols. (2) and (4)), the correlation between our gap measures and the maximum FAR values increases. In addition, the coefficients of the other variables are, for the most part, not significant. The only other land use policies for which we find a significant effect are urban growth boundaries and minimum plot size. Measures aimed at controlling sprawl (GB, UGB) have negative coefficients, implying that the gaps are lower when cities cannot build "out" and have to build "up." Zoning and government planning are then associated with higher gaps. Lastly, the existence of a minimum plot size implies lower gaps since it facilitates the construction of taller structures. More generally, while the gaps are strongly correlated with building-height restrictions, they seem to capture a broader set of land-use policies.

Second, countries that stringently restrict the height of their buildings could compensate their residents by having more lenient regulations in other dimensions, for example control sprawl less. If such compensation mechanisms indeed exist, we should observe that low-FAR countries have less stringent policies in other dimensions, hence finding a strong positive correlation between the maximum FAR variable and the other measures of land use regulations. However, the correlation is weakly negative in the majority of cases (col. (5)). Thus, low-FAR countries are *not less likely* to restrict sprawl through urban containment policies (-0.15 for GB and -0.12 for UGB).

6. The Gaps and other Urban Outcomes

Tables 7 and 8 present correlations between the gaps and measures of housing prices, sprawl, congestion, and pollution. We always control for national per capita income. For a given income level, we thus examine how the gaps might have economic and environmental consequences. Controlling for income also ensures that we do not simply compare rich and poor countries. One important caveat is that the results are not causal.

Housing Prices. World Bank (2019a) reports the price level of broad consumption categories in 2011 (relative to the world = 100). In col. (1) of Table 7, we regress the price level of housing on the gaps while controlling for log nominal (national) per capita GDP (2010) (World Bank, 2019b) and using as weights urban population (2010) (N = 147). We control for nominal GDP because higher housing prices would be captured by PPP adjustments given how typically high housing expenditure shares are.[23] A unitary decrease in the H-based gap is associated with 4% lower housing prices. The magnitude of the effect is large: A one standard deviation increase in the gap (a value of 2) is associated with a 0.15 standard deviation increase in the price level.

[23] As discussed in Section 4, the urban population weights are important to give less weight to smaller countries as such countries might be more likely to mechanically have larger percentage gaps. We also need to make the analysis representative of the global distribution of urban populations.

Table 7
Gaps and economic outcomes, country-level analysis.

Dep. Var.:	World Bank '11		Global Prop. Guide'19		Knoll et al 2017		Share
	Price Level (100)		Log Hous	Price-	Real Hous. Price t		Public
	Housing	Transp.	Price ($)	to-Rent	Long-Diff	Short-Diff	Housing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$Gaps_H$	3.99***	-1.50	0.24***	3.50***	28.78*	14.78**	0.08
	[1.33]	[1.34]	[0.04]	[0.84]	[14.38]	[6.48]	[0.78]
$Gaps_{UMH}$	3.32***	-0.67	0.18***	2.78***	13.27	8.33	0.48
	[1.19]	[1.30]	[0.04]	[0.60]	[16.62]	[6.01]	[0.58]
Country, Year FE	N	N	N	N	Y	Y	N
Controls, Weights	Y	Y	Y	Y	N	N	Y
Observations	147	147	72	70	28	83	48

Dep. Var.:	Col. (8)-(12): Log Total Urban Land Area (Km) in...					Log Particulate Matter Level (PM)	
	World	Col. (9)-(10): GHS t		Ctrl: Built-Up Area t			
	Bank '11	Long-Diff	Short-Diff	Long-Diff	Short-Diff	10 ('10)	2.5 ('17)
	(8)	(9)	(10)	(11)	(12)	(13)	(14)
$Gaps_H$	0.19***	0.05**	0.03**	0.04*	0.02*	0.05**	0.07**
	[0.03]	[0.02]	[0.01]	[0.02]	[0.01]	[0.02]	[0.03]
$Gaps_{UMH}$	0.22***	0.06***	0.03**	0.05*	0.02*	0.05**	0.08***
	[0.04]	[0.02]	[0.01]	[0.03]	[0.01]	[0.02]	[0.02]
Country, Year FE	N	Y	Y	Y	Y	N	N
Controls, Weights	Y	Y	Y	Y	Y	Y	Y
Observations	126	262	524	262	524	146	156

Notes: Columns (1)-(4), (7)-(8), and (13)-(14): See text for details on the specifications. Columns (5)-(6): Panel regressions for 14 countries in 1960-2010 (1960, 1970, 1980, 1990, 2000, 2010). Long-diff: We consider the first and last years only. Columns (9)-(12): Panel regressions for 131 countries in 1975-2015 (1975, 1990, 2000, 2015). Long-diff: We consider the first and last years only. Robust SEs (clustered at the country level when using panel regressions in columns (5)-(6) and (9)-(12)): * p<0.10, ** p<0.05, *** p<0.01.

Recall from above that such a unitary decrease corresponds to a 100% increase in actual height density.[24]

Countries with larger gaps could compensate their urban residents by subsidizing commuting, for example via public investments in urban transportation infrastructure. In column (2), we use the same specification but regress the price level of transportation on the gaps while also controlling for the price level of housing (N = 147). While negative coefficients are observed, the point estimates are not significant. Thus, the higher housing prices do not appear to be compensated by cheaper transportation.

Another data set on global property prices is the *World's most expensive cities* list provided by GPG (2019). For the largest city in 75 countries, the list shows selling prices per square meter as well as the price-to-rent ratio (PRR). Typically, a high PRR suggests that the costs of housing will increase in the future. The two measures are available for 72 and 70 countries in our sample, respectively. The selling price ranges from 700 USD per square meter in Dar-es-Salaam to 30,000 USD per sq m in Hong Kong, while the PRR ranges from about 10 in Kingston to 50 in Vienna. In columns (3)-(4), we regress these measures on the gaps while controlling for log nominal (national) per capita GDP (2017), log city population size (2015), and using as weights urban population (2017). The gaps strongly correlate with current and future housing prices (captured by PRR). A unitary decrease in the H-based gap is associated with 24% lower housing prices (col. (3)). A one standard deviation increase in the gap is then associated with a 0.56 standard deviation increase in prices. We also find large coefficients for future housing prices (col. (4)).

Knoll et al. (2017) show that housing prices have increased faster than overall prices for 14 countries post-1950. Fig. 6 shows that the evolution of the total km gap of the 14 countries follows the evolution of mean real house prices in their sample. Cols. (5)-(6) show using panel regressions for the 14 countries (1960-2010) the strong correlation between the gaps and real house prices (country and year fixed effects included; standard errors clustered at the country level). In col. (5), we keep 1960 and 2010 to study long-difference correlations. A unitary decrease in the H-based gap is associated with 29% lower prices. Alternatively, a one standard deviation increase in the gap is associated with a 0.60 standard deviation increase in prices. The short-difference correlation (col. (6)), estimated using all available years, is halved, implying that gaps might have effects in the following decades. Lastly, proportionate reduction of error analysis suggests the H-based gaps might have explained 22% of the global house price boom.

Next, although expensive high-gap countries might exclude the urban poor unless housing is subsidized, the share of housing that is publicly provided (HOFINET, 2020) is not correlated with the gaps (col. (7); using the specification of cols. (1)-(2); N = 47).

Finally, given the lack of global historical data on non-tall buildings, we are unable to assess whether higher-gap countries have smaller gaps in this other dimension. However, the fact that housing prices are higher in higher-gap countries suggests that any compensation via the non-tall sector may not be sufficient. Below in Section 7, we also discuss why tall building construction might be a good proxy for overall construction.

To summarize, high-gap countries do not appear to "compensate" their urban residents by subsidizing transportation or directly providing housing to the urban poor. In the largest cities where constraints on vertical development are more likely to be binding, we found that housing prices are about 20% higher. With a 30% housing expenditure share, this would correspond to a 6 percentage decrease in purchasing power. Could tourism receipts justify such effects? First recall that we did not find any correlation between the gaps and tourism receipts. Second, in the most tourism-oriented high-gap nations, for example France and Italy, the GDP contribution of tourism is about 2.5% (which also includes tourism outside the largest cities). Thus, tourism might not be

[24] Occupancy may vary with the gaps in that we would expect high-gap countries to have lower vacancy rates, other things equal. Using office vacancy data from Colliers International (2006), we find evidence for lower vacancy rates in higher-gap countries. The correlations are mostly not significant, however (Web Appx. Table A3). Also, changes in vacancy rates should eventually result in price changes, which are likely better long-run measures.

Table 8
Gaps and economic outcomes, city-level analysis.

Dep. Var.:	Log Sum of Heights in Year t				Log Area in Year t		Congestion Level 2017	Log PM'17	
			Central Areas	Periph. Areas				10	2.5
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: H Gaps:									
1(100-500K)*GAP	-0.04	-0.03	-0.03	0.00	0.01	0.01	1.79**	0.04	0.04
	[0.03]	[0.02]	[0.02]	[0.00]	[0.01]	[0.02]	[0.73]	[0.03]	[0.03]
1(500-1000K)*GAP	-0.25*	-0.11	-0.08	-0.02	0.00	0.00	2.71*	0.09**	0.08*
	[0.14]	[0.17]	[0.15]	[0.03]	[0.03]	[0.02]	[1.49]	[0.04]	[0.04]
1(1000K+)*GAP	-0.73**	-0.56**	-0.54**	-0.15*	-0.02	0.00	1.48	0.13**	0.14***
	[0.31]	[0.25]	[0.24]	[0.09]	[0.02]	[0.01]	[1.13]	[0.05]	[0.05]
Panel B: UMH Gaps:									
1(100-500K)*Gap	-0.02	-0.02	-0.02	0.00	0.05	0.04**	1.14***	0.02	0.03*
	[0.04]	[0.01]	[0.01]	[0.00]	[0.03]	[0.02]	[0.37]	[0.02]	[0.02]
1(500-1000K)*Gap	-0.17	-0.13	-0.10	0.01	0.09	0.05	1.43*	0.06**	0.06***
	[0.32]	[0.12]	[0.12]	[0.04]	[0.06]	[0.03]	[0.77]	[0.02]	[0.02]
1(1000K+)*Gap	-0.85	-0.68***	-0.68***	-0.01	-0.02	0.01**	1.04	0.07**	0.08***
	[0.56]	[0.17]	[0.16]	[0.13]	[0.05]	[0.01]	[0.64]	[0.03]	[0.03]
City, Year FE	Y	Y	Y	Y	Y	Y	N	N	N
Country-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	23,438	46,876	46,876	46,876	17,040	34,181	391	1,473	1,473
Number of Cities	11,719	11,719	11,719	11,719	11,719	11,719	391	1,473	1,473
Number of Years	2	4	4	4	2	4	1	1	1

Notes: Columns (1)-(6): Panel regressions for 11,719 50K+ GHS urban agglomerations in 1975-2015 (1975, 1990, 2000, 2015). We interact the country-level gaps with three city population category dummies defined in 1975 (50-100K is the omitted category). Columns (7)-(9): See text for details on the specifications. Robust SEs clustered at the country level: * p<0.10, ** p<0.05, *** p<0.01.

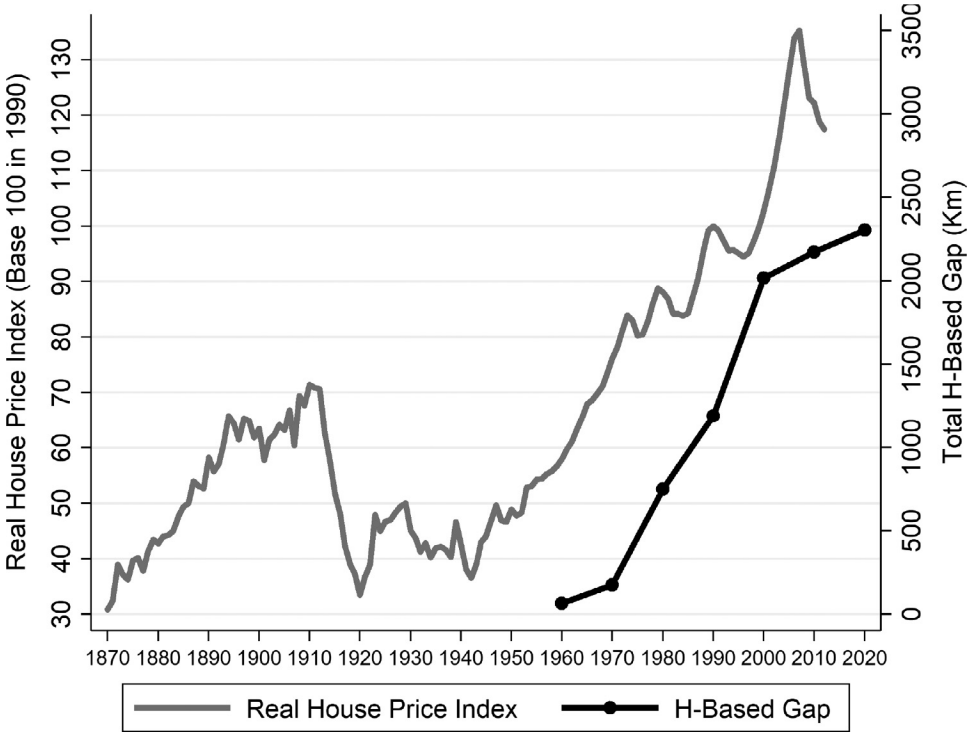


Fig. 6. HOUSE PRICES & BUILDING-HEIGHT GAPS, 14 COUNTRIES, 1870-2020. *Notes:* This figure shows that real house prices have dramatically increased in developed countries since the 1950s (we use the data from Knoll et al. (2017)). It also shows that the estimated total km (H-based) gap of the 14 countries has increased since 1960. More precisely, Knoll et al. (2017) reports a real house price index (base 100 in 1990) for 14 OECD countries annually from 1870 to 2012. We then obtain the average real house price index for the 14 countries in each year using the population of each country as weights.

the only factor why tall building construction is constrained in high-gap countries.

Urban Land Expansion: Country-Level Results. If cities cannot expand vertically, they may need to expand horizontally. Therefore, for a given urban population and a given per capita income level, we expect countries with larger gaps to use more urban land. However, since housing prices are overall higher, it may be that horizontal expansion is not enough to "compensate" for the lack of vertical expansion.

In col. (8) of Table 7,we regress total urban land area (2011) (World Bank, 2019b) on the gaps (N = 125). We control for log urban population (2010), log nominal (national) per capita GDP (2010) (since higher incomes increase housing/land consumption and imply better commuting technologies), and log nominal agricultural land rent (2010) (since a higher land rent constrains land expansion). We use urban population weights (2010). Land expansion is positively correlated with the gaps. A unitary increase in the H-based gap is associated with cities consuming 19% more land. A one standard deviation increase in

the gap is then associated with a 0.23 standard deviation increase in urban land area.

Moreover, we use the *Global Human Settlement* (GHS) database of European Commission (2018) to obtain for 131 countries in 1975, in 1990, in 2000 and in 2015 the total population, total land area, and total built-up area (thus excluding open spaces) of all (11,719) 50K+ urban agglomerations today. In cols. (9)-(10), we use as the dependent variable the log of total city land area in t while adding country and year fixed effects and controlling for log total city population, log nominal (national) per capita GDP, and log nominal agricultural land rent in t, with the variable of interest being the gap in t (standard errors clustered at the country level). By restricting our panel analysis to the years 1975 and 2015 (N = 262), we capture how the gaps correlate with the long-difference change in urban land per capita. A unitary decrease in the H-based gap is associated with urban areas consuming 5% less land. A one standard deviation in the H-based gap is then associated with a 0.07 standard deviation increase in urban total land area. With the full panel (N = 524), elasticities are halved (col. (10)).

Simultaneously controlling for log total *built-up* area (t), col. (11)-(12) further shows that the gaps correlate with sprawl. Indeed, since the outcome is total land area, these results suggest that cities in high-gap countries use more land per built-up area, i.e., have more open areas and are less compact generally (col. 13-14 are discussed below).

Theoretically, by restricting housing supply, constraints on vertical development raise the price per unit of housing throughout the city while causing the urban footprint to expand. Residents experience a combination of higher housing prices and longer commutes. For the resident at the edge of the city, the welfare loss comes entirely from a longer commute. With utilities equalized within the city, the welfare loss for each resident equals the increase in commuting cost for the edge resident (Bertaud and Brueckner, 2005). In Web Appx. Section B, we describe how we can use our results on urban land expansion and various assumptions to estimate this increase in commuting cost. In particular, we show that the implied costs from a one-standard-deviation increase in the gaps (≈ 2) – which corresponds to the world gap – is possibly 0.7-1.0 percent of total urban income. Crudely adding the cost of particulate matter pollution (0.4-0.8% of world GDP, see details below), we obtain 1.1-1.8% of world GDP. Our back-of-the-envelope calculations thus suggest that the gaps might have major economic costs globally.

Urban Land Expansion: City-Level Results. While the previous conclusions are all derived at the country level, we can combine the country-level gaps with city-level information to generate some additional insights, as follows. We take advantage of the fact that the GHS database also reports estimates of total land area for all 11,719 agglomerations circa 1975, 1990, 2000 and 2015. More precisely, we ask whether the height difference between larger and smaller cities is reduced in countries where gaps are high.

For the years 1975 and 2015, we run city-level panel regressions where the dependent variable is the log sum of heights (t) and the variables of interest are the country gaps (t) interacted with three dummies for whether the city has 100-500K, 500-1,000K and 1,000K+ inhabitants in 1975, respectively. We use 1975 because post-1975 changes in city populations are endogenous to post-1975 changes in the country gaps. We include city fixed effects, country-year fixed effects, and cluster standard errors at the country level. Moreover, we control for log urban per capita GDP (PPP, 1990 constant international $) interacted with the dummies to capture how changes in the gaps occur in larger cities rather than the fact that gaps are becoming larger in countries with richer urban areas.[25] Lastly, we use urban population weights (t) to minimize the low-stock issue and make the analysis representative of the global distribution of urban populations.

Table 8 reports the results. The effects of the country gaps are particularly visible for larger cities (col. (1)), thus suggesting that they are associated with abnormally constrained big cities. If we use the full panel (1975, 1990, 2000, 2015) to focus on short-term correlations, the point estimates are reduced, but significant above 1,000K (col. (2)).

As such, we do not use our city-level data to estimate the country gaps themselves but, instead, to show that the country gaps are driven by the largest cities in each country (conditional on city fixed effects that absorb city-level time-invariant factors).

Next, using the full panel specification, we confirm that the gaps are more constraining in the central areas of urban agglomerations (col. (3)). We then test if such gaps are compensated by vertical development in peripheral areas. For example, most tall buildings in the Paris and Washington DC agglomerations are located in the peripheral La Défense and Arlington areas, respectively. We re-run the same regression using the log sum of heights in peripheral areas and find, however, that the coefficients of the gaps interacted with the city dummies are nil or negative, and not positive (col. (4)). Thus, it appears that high-gap countries abnormally constrain *all* areas of big agglomerations.

If central area gaps are not compensated by vertical development in peripheral areas, larger cities may expand beyond their initial boundaries. We test for such an effect using the city-level panel specifications except that the dependent variable is now log city total area (cols. (5)-(6)). No correlation is found for the H-based gaps. These regressions compare relative land expansion patterns for different class sizes of cities (50-55K is the omitted category) whereas cols. (9)-(10) of Table 7 examined the total expansion of all urban areas. Since urban land expansion is correlated with the gaps at the country level, *all* class sizes of cities may be expanding spatially due to binding gaps in the largest cities.

Finally, as cities sprawl they may become more congested, especially if workers have to rely on motorized vehicles for their commute. We test that notion next.

Congestion: City-Level Results. Traffic congestion is available for 391 50K+ cities today (TomTom, 2019). The measure indicates by how many percentage points commuting times increase during rush hours relative to non-rush hours.[26] Congestion strongly increases with log population size (N = 391; with country FE; coef. = 3.7***; adj. $R^2 = 0.75$). We then examine how this relationship is affected by the country gaps. We regress congestion on the gaps again interacted with the three city-size dummies. We include country fixed effects and log (national) per capita income interacted with the population category dummies, and use urban populations (2020) as weights (standard errors clustered at the country level). Larger cities are disproportionately more congested than 50-100K cities in higher-gap countries (see col. (7) of Table 8). Finally, knowing the population share of each group of cities, we compute the average effect across the three groups, equal to 1.55 (significant at 10%) for the H-based gaps. Thus, a unit increase in the H-based gap is associated with 1.5% more congestion. The magnitude is large: a one standard deviation in the H-based gap possibly raises congestion by 3%. Finally, the correlation is halved and insignificant when controlling for sprawl 1975-2015 (not shown).

Pollution: Country-Level Results. With sprawl and road congestion, pollution may also increase, implying that gaps might have environmental consequences. Air pollution in cities consists of gases – mostly carbon dioxide (CO_2) and nitrogen oxides (NO_x) – and particulate matter (PM) measured by their size, such as 10 and 2.5 micrometers. CO_2, NO_x and PM have health effects. CO_2 and NO_x also contribute to global warming. Unfortunately, ground-based measures of CO_2 and NO_x are not available for enough urban areas across the world. However, there is data on PM10 and PM2.5. In columns (13)-(14) of Table 7, the dependent variables are the log levels of PM 10 (2010) and PM 2.5 (2017)

[25] For cities a belonging to population categories p, countries c and years t, the model is: $\text{LHEIGHT}_{a,c,t} = \alpha + \Sigma_p \beta_p \mathbb{1}(\text{POP}_{a,c,1975} = p) \times \text{GAP}_{c,t} + \Sigma_p \gamma_p \mathbb{1}(\text{POP}_{a,c,1975} = p) \times \text{LUPCGDP}_{c,t} + \theta_a + \kappa_{c,t} + \mu_{a,c,t}$.

[26] TomTom constructs the measure using its own data on the travel patterns of 600 million drivers (accessed 02-28-2020: https://www.tomtom.com/engb/traffic-index/).

in more populated areas, respectively (World Bank, 2019b).[27] We control for log (national) per capita GDP and log urban population (2010 or 2020), and use urban populations (2010 or 2020) as weights. A one point increase in the gap is associated with 0.05-0.08% more pollution. A one standard deviation increase in the gaps is then associated with a 0.05-0.08 standard deviation increase in PM.

Pollution: City-Level Results. For 1,473 GHS agglomerations, we obtain from WHO (2019) the average levels of PM10 and PM2.5 in 2008-2017. Given the same specification as for congestion, gaps are associated with increased pollution in the largest cities (see col. (8)-(9) of Table 7). Knowing the population share of each group of cities, we obtain the average coefficient across the three groups, 0.04^{*}-0.07^{**} for PM10 and 0.05^{***}-0.08^{***} for PM2.5 when considering H-based gaps. Thus, a one point increase in the H-based gap possibly raises pollution by 4-8%. Alternatively, a one standard deviation in the H-based gaps (= 2) may raise pollution by 8-16%. Now, the cost of pollution is 4.8% of world GDP (World Bank, 2016). Thus, a world gap increase of 2 might reduce world GDP by 0.4-0.8%.

Finally, whether for the country-level or city-level regressions, if we control for log land area and log city population in 2015, and their squares in case congestion varies non-linearly with sprawl, the correlations of the gaps with pollution are strongly reduced (Web Appx. Table A4). Most of the pollution results thus seem explained by sprawl.

7. Robustness Checks

We now address various concerns related to: (i) the selection of the benchmark countries; (ii) causality; (iii) the baseline specification; and (iv) measurement error.

Sampling Checks. The results depend on the selected benchmark countries. While our selection method has clear limitations, there is unfortunately no objective process that might help us distinguish more vs. less supply-elastic countries, hence the need to assess how robust our results are to the choice of the benchmark set.

First, a Google search supports the validity of the process by which we select supply-elastic countries, as follows. For 61 UM-H countries, the correlation between the selection residuals and the number of Google search results for the country name & “cities” & “skyscrapers” is 0.61 (conditioning on the respective numbers of search results for the country name & “cities” and the country name). The correlation is 0.72 if we also control for whether English is an official language and the numbers of famous 20th and 21st century architects from the country (Wikipedia, 2020b), as some countries have renowned architects but few skyscrapers (e.g., Italy with Renzo Piano). With search results in the country’s language, the correlation becomes 0.90.

Next, our benchmark set does not include the U.S., Japan, and China, three countries considered as having many tall buildings. However, that is not necessarily the case once one accounts for the size of their urban population and their economic conditions. While some of their cities have impressive skylines, these countries also have many large cities with restricted FARs. Yet, since the three countries are close to being included in the benchmark set based on the selection residuals, we verify results hold if we include them in the benchmark set, whether the U.S. only, the U.S. and Japan or China, or the three countries altogether (Web Appx. Table A5). When doing so, the country rankings are mostly unchanged (the correlation between the various gap measures is above 0.90).

There could also be a concern that the benchmark set over-represents high-income observations and that we apply their coefficients to non-benchmark countries with lower incomes. However, the coefficients are estimated for the period 1950-2020 when many of today’s UMH countries had low income levels (two fifths of the 14 countries x 8 years = 112 observations have income levels corresponding to the LLM category). Nonetheless, if we consider democratic countries whose residual is above the 75th percentile value without restricting the sample to UMH countries, this adds four LM countries. When also using these as benchmark countries, or Indonesia and the Philippines only (Honduras and Mongolia were non-democratic for most of the period), we get similar coefficients as for the UMH set (Web Appx. Table A1). The gaps that we obtain then strongly correlate with the gaps obtained with the UMH/H set (correlation = 0.76-1.00).[28]

A related question is whether the benchmark set is sufficiently representative of the rest of the world in terms of total land area. Classical measurement error in land rent should lead us to under-estimate its coefficient and the gaps. But land rent at the edge of cities may be mismeasured for larger countries. Our benchmark sets include three large countries (Australia, Brazil, Canada) and two small countries (Hong Kong and Singapore). If the benchmark set has too many large (or small) countries relative to the rest of the world, the estimated coefficient may affect our results.

However, land rent is likely under-estimated in larger countries (since average land rent captures more rural areas away from cities) whereas the gaps may appear larger there since development should take place more horizontally due to land availability. If anything, this should also lead to a downward bias of the coefficient/gaps. Second, we use the *OECD Regional Database* of OECD (2021) to examine for 505 regions in 36 countries (incl. 6 non-OECD countries) how much of a good proxy is national agricultural land rent for agricultural land rent in more urban regions. More precisely, the database reports agricultural GDP and total land area for each region c. 2010 (data unavailable before 2004).

For 264 regions with a metro area (a large metro area), the correlation between regional and national rent is 0.87 (0.89) in countries below the median land area in the sample and 0.80 (0.82) in countries above. While the correlation is lower in larger countries, it is still very high. Global differences in land rent mostly arise between countries, not regions within countries. Third, Kolmogorov-Smirnov tests suggest that the Kernel distribution of total land areas is not significantly different between the benchmark and non-benchmark countries for both sets (not shown). Fourth, the land rent coefficient barely changes if we exclude the three large countries or the two small countries (Web Appx. Table A5). Lastly, the coefficient of income is mostly unchanged if we omit land rent (Ibid.). In all cases, the country rankings remain similar (not shown).

Finally, results hold if we drop each country one by one, so the coefficients and country rankings are not affected by any particular country in the benchmark set (not shown). Nonetheless, we acknowledge that there is no obvious way to select the laissez-faire countries, and our results and estimated country gaps should thus be taken with caution.

Causality. The gaps are over-estimated if the coefficients of income and land rent in Table 1 are upward biased. A downward bias would make us under-estimate the gaps, which is less consequential. However, the bias would most likely affect the levels of the gaps, not country rankings. Despite different coefficients of income and land rent, we found that the correlation between the H- and UMH-based rankings was above 0.85.

[27] PM 10 is measured for urban areas above 100,000 inhabitants only. The mean level exposure of a nation’s population to PM 2.5 air pollution is then computed by using the PM 2.5 level and population of different areas in each country. As such, the measure overly represents populated urban areas.

[28] Alternatively, if we re-add the two most democratic LM countries – Indonesia and the Philippines – and allow the coefficients to change across different levels of the explanatory variables, we find that the coefficients of urban income and land rent are remarkably stable (Web Appx. Table A6). Thus, if the income coefficient is larger for the H set (as of today) than for the UMH LM-UMH set or the LM-UMH sets (Ibid.), it is not because of their observations with higher incomes but because they must indeed be more laissez-faire.

Next, such bias would be a consequence of the correlation between the explanatory variables and the regression error term. This correlation could arise either from omitted variables or from reverse causality. In the omitted variable case, a country's commitment to free-market principles may raise both its urban income level and the height of its buildings, leading to an upward bias. Alternatively, effective transit systems may influence both incomes and building heights. While use of country fixed effects mitigates the effect of such unobservables to some extent, bias is still a concern. Examples of reverse causality include a positive feedback effect from commercial buildings to incomes operating through agglomeration economies. Alternatively, the supply-increasing effects of taller buildings may reduce housing prices enough to attract low-income consumers to cities, generating a negative feedback effect on income. Another example might be negative feedback (via reduced sprawl) from heights to agricultural land values, in which a more compact city relieves price pressure on surrounding farmland.

We do not believe that there exists an identification strategy that would fully allay these concerns; an instrument that would explain incomes without impacting heights is hard to find. We thus discuss another series of results that gives us greater confidence in the coefficients. To this end, Web Appx. Table A7 shows results generally hold if:

(i) We include continent- or World Bank region-year fixed effects, in order to capture regional economic, institutional and cultural drivers of tall building construction that may affect building heights, urban incomes and agricultural land rent;

(ii) We include country-specific linear trends or even country-specific non-linear trends, i.e., country dummies interacted with the year and the square of the year. In that case, identification comes from swift or very swift (and possibly more exogenous) growth (or deceleration) within countries, i.e., deviations from country trends.

(iii) We capture commuting costs by controlling for whether there is a subway and the logs of the number of subway lines and stations in the country in year t (source: Gonzalez-Navarro and Turner (2018); Gendron-Carrier et al. (2018)) as well as the percentage of country roads (including non-urban roads) that is paved during the period 1990-2017 interacted with a linear year trend (source: World Bank (2019b)). The coefficient of income increases. This change makes sense if richer countries have better transportation infrastructure, and if lower commuting costs reduce the need to build up. Then, a negative correlation between the error term and income arises, creating downward bias in the income coefficient, which is reduced by controlling for commuting infrastructure. Thus, had we better controls for commuting, the income coefficient would likely increase.

(iv) We add leads of the main explanatory variables to address possible reverse causality, with the variables defined as t + 10. The leads have no effects. Tall buildings are not built in anticipation of future income growth (at least not ten years in advance).

(v) We control for time-invariant geographical and other factors interacted with year fixed effects (thus allowing their effects to vary over time) in case they constrain tall building construction and affect income/land rent. These factors are the logged (population-weighted) average earthquake risk and bedrock depth in urban areas, the share of the population living in mountainous areas, and the price level of construction in 2011 (see Section 5 for details on the sources). Note that results also hold if we control for the other possible determinants of the gaps considered in Table 4 (not shown).

Finally, using the 11 different regressions in Web Appx. Table A7, we can generate 11 different gap rankings for the sample countries. For $11 \times 10 \div 2 = 55$ pairs of rankings, the mean and 5th percentile correlations are 0.96, and 0.90, respectively. Rankings are thus largely insensitive to the exact magnitudes of coefficients used.

Measurement Error in Building Heights. The dependent variable is log urban height density, which is the sum of heights (for buildings above 80 or the top 10 buildings) divided by urban population.[29] Classical measurement error in dependent variables only affects precision. However, measurement error could be non-classical.

To compare results, we collected data from a second source, Emporis (2019), another global provider of building information. Note that Emporis (2019) claims to capture all *high-rise buildings*, which they define as buildings above 35 meters (about 9 floors). They then classify as *skyscrapers* buildings above 100 meters. Finally, they use the number of floors of each 35m+ building to compute for each city a Skyline Index. We do not have access to their raw data but their website reports useful information for the 100 top cities in the world.[30] For 90 of these cities also in our data, and using as weights the sum of heights in our data in order to focus on the cities with the most tall buildings, the correlation between the log of their number of skyscrapers and the log of our own number of buildings above 100 meters is 0.90. Next, the correlation between the log of their Skyline index and the log of their number of skyscrapers is 0.83. The correlation of their Skyline Index with our own reconstructed index (using our data and their formula) is 0.79. Thus, our measure of urban height density is a good proxy for 35m+ buildings.

Now, is our measure also a good proxy for structures below 35m, whether low-rise (four plus one) buildings or houses? Based on Emporis, which also reports the number of low-rise buildings for seven North American cities, the (mostly 80-m+) buildings in our data account for between half and two thirds of total heights including low-rises. In addition, for each building, we know the main material used. While it was steel around 1950, the use of concrete has dramatically increased over time, reaching 90% in the 2000s (Fig. 7). Next, we obtained from the *Minerals Yearbooks* of USGS and for 144 countries and each decade from 1950 the total production of cement – the main ingredient of concrete – which we use as a good proxy for cement consumption. The correlation between decadal tall building construction and decadal cement use is high, at 0.77 (N = 870). Adding country and year FE, we obtain a correlation of 0.80 (0.99 with urban population as weights). Thus, tall building construction is a good proxy for overall construction.

Next, one could argue that taller skyscrapers are better measured than shorter high-rise buildings, because they stand out more. Among the buildings in our data, the height of the 25th percentile, median, and mean is 100, 125 and 135 meters, respectively. The results hold if we restrict our analysis to buildings above such thresholds, or even 150 meters, which is about the size of the tallest buildings in sub-Saharan Africa outside South Africa (Web Appx. Table A5). While there might be concerns that tall buildings are undercounted in poorer countries, we personally doubt that the CTBUH would not systemically report buildings of such sizes in such countries, since there are so few of them. Thus, restricting the sample to higher thresholds should take care of non-classical measurement error in building heights. We then verify that the country rankings are mostly unchanged when doing so (not shown). Finally, our results hold if we drop heights that are imputed based on the number of floors (Ibid.).

Lag of the Dependent Variable. With panel regressions, including a lagged dependent variable might introduce dynamic panel bias (Nickell, 1981). However, Web Appx. Table A5 shows that the coefficient of GDP significantly increases when omitting the lagged dependent variable, which would lead to even higher gaps (country rankings are also little affected; not shown). Note that we keep the lagged dependent variable in the baseline model because of the need to control for the durability of tall buildings.

[29] Results hold if we use alternative ways to handle the zeros, i.e. country-years without buildings above 80 m: An inverse hyperbolic sin regression or a negative binomial regression (Web Appx. Table A8).

[30] Accessed on 12-11-2019: https://www.emporis.com/statistics/skyline-ranking.

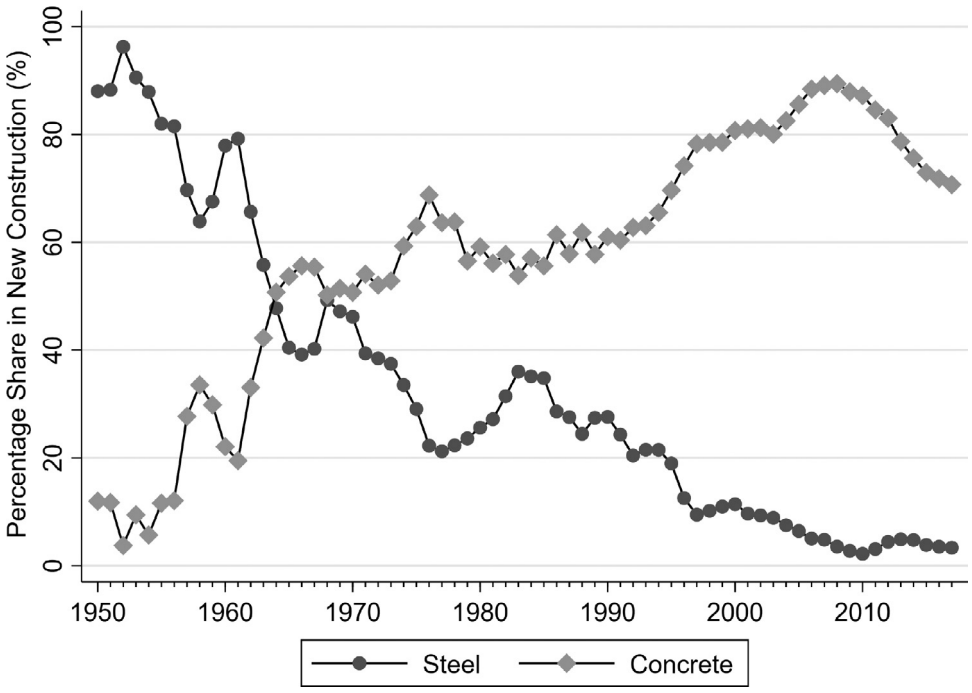


Fig. 7. STEEL VS. CONCRETE IN NEW CONSTRUCTION, 1950-2017. *Notes:* This figure shows for each year the share of new construction (weighted by building heights) that comes from buildings whose main material is steel vs. concrete. These shares are obtained using available information for 10,809 out of the 16,369 buildings in our data. We report two-year moving averages.

Lastly, countries that may have developed their downtown cores to their optimal level should not build more tall buildings. Therefore, Web Appx. Table A5 shows results hold if we include the square of the lagged log urban height density in case the ability of a country's cities to redevelop "higher" is decreasing with the existing urban height density. We also try including the cube and fourth power of the variable. Country rankings are little affected (not shown). Indeed, while a few cities have "crowded" central areas, many of their buildings could still be redeveloped higher, and other cities in the same countries do not necessarily have that many tall buildings.

U.S. State Analysis. In Web Appx. Section C, we perform a similar gap analysis using decadal U.S. state-level data from 1929 to 2017, which offers another opportunity to validate our methodology. Using the same process as at the country level, we first identify states with higher residuals today, i.e., states that may be more supply-elastic for various reasons. We then run panel regressions that relate building heights to a state's income, agricultural rent, and past building stock. As expected, we find higher coefficients of income and land rent when restricting the sample to the benchmark states (e.g., Illinois and New York). We then use the data and these coefficients to obtain the gaps for each state, finding that California might account for 48-61% of the U.S. gap. We then show that the gaps strongly correlate with the supply elasticities of Saiz (2010) (which also include geographical constraints) and well-known measures of land-use regulations from Gyourko et al. (2008) and Saks (2008). Finally, if we use the U.S. state coefficient estimates to obtain predicted heights and the gaps for all countries in the world, the coefficient of correlation between the H-based gap measure and the U.S. state-based gap measures is 0.74-0.76 (using urban population as weights). If we use the UMH set instead, the correlation is also high, at 0.72-0.89. This increases our confidence in the country-level benchmark regression as a tool for computing building-height gaps across the world.

8. Conclusion

Using a new data set on nearly all tall buildings on the planet, this paper has constructed a measure of building-height gaps around the world. In particular, we first obtain parameters from a regression of building heights on income and agricultural rents for a benchmark set of countries that appear to have higher supply elasticities. Using regression results for these benchmark countries, we create building-height gaps for countries around the world. We find that, under laissez-faire conditions, the world would likely have twice as many tall buildings (about 6,000 Empire State Buildings) as it does currently.

We show that the gaps are relatively larger for richer countries, especially European countries. Globally, the gaps are larger for office towers, due to various world regions having disproportionately large residential tall-building stocks (e.g., the MENA countries). However, in developed countries, the gaps are relatively larger for residential buildings.

Conditional on income, the gaps are strongly and positively correlated with housing prices, sprawl, road congestion, and pollution, implying that they might have important global economic and environmental consequences. In particular, such gaps might explain one fifth of the global house price boom observed since 1950. The gaps then appear to constrain the growth of large cities and their central areas. We do not find that countries with larger building-height gaps "compensate" their urban residents: (i) by having more lenient height restrictions outside their cities' central areas; (ii) by having more lenient sprawl policies; (iii) by subsidizing transport; or (iv) by providing public housing.

Regarding the nature of these gaps, we find no evidence that they are caused by geological, topographical, or seismic conditions, or differences in construction costs. However, countries with more "historical" cities have higher gaps. Evidently, the gaps in these countries arise from regulations designed to protect valuable historical urban areas, motivated by cultural considerations or a desire to foster tourism. However, countries may also adopt stringent regulations for reasons unrelated to historical preservation.

Additionally, we have worked through a series of robustness checks, estimating the gaps with different specifications, and estimating gaps at the city level and U.S. state level. Across all these tests, we find strong correlations of our country-level gap with other gap measures. The results suggest that our gap measure is an useful index of tall building supply across the world, and which emerge from various local and national frictions.

CRediT authorship contribution statement

Remi Jedwab: Conceptualization, Methodology, Formal analysis, Validation, Investigation, Writing – original draft, Funding acquisition. **Jason Barr:** Methodology, Investigation, Resources, Writing – original

draft, Writing – review & editing. **Jan K. Brueckner:** Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jue.2022.103507

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